Guana Tolomato Matanzas National Estuarine Research Reserve

SEACAR Water Quality Analysis

Last compiled on 05 May, 2026

Contents

Indicators

Nutrients

Total Nitrogen - Discrete

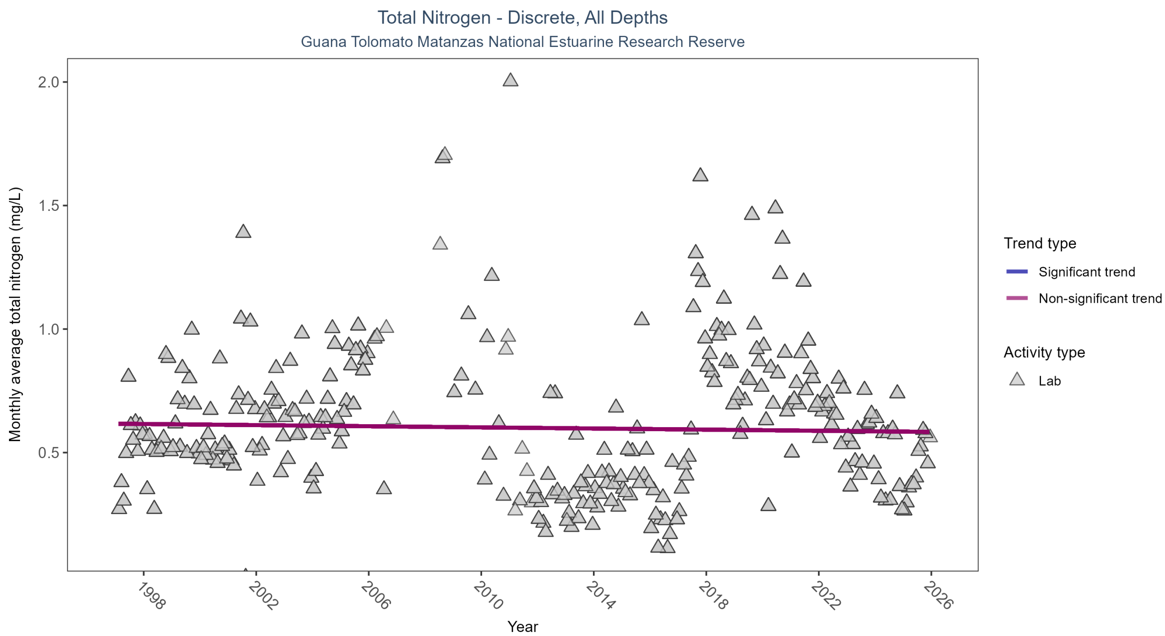

Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	3682	28	1997 - 2025	0.5465	-0.02361	0.6169	-0.00114	0.5711

Total nitrogen showed no detectable trend between 1997 and 2025.

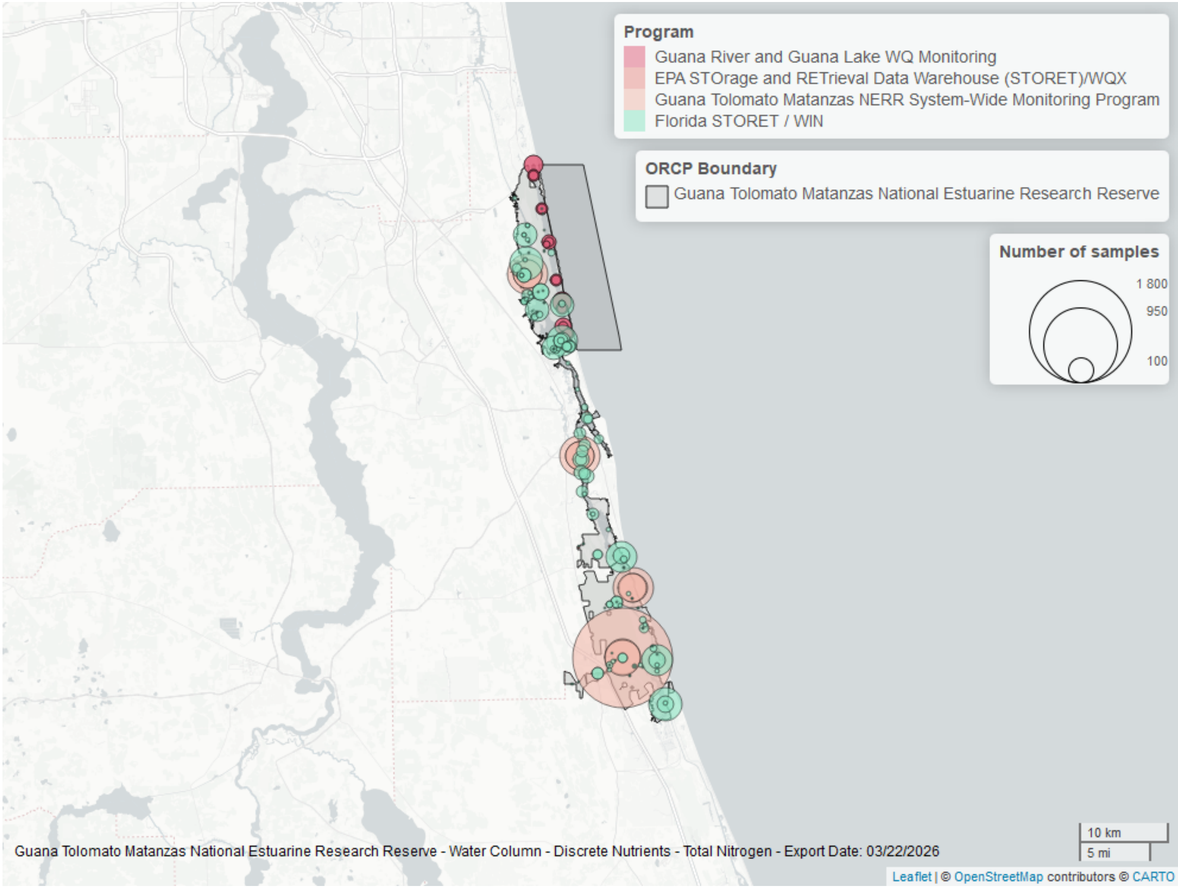

Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

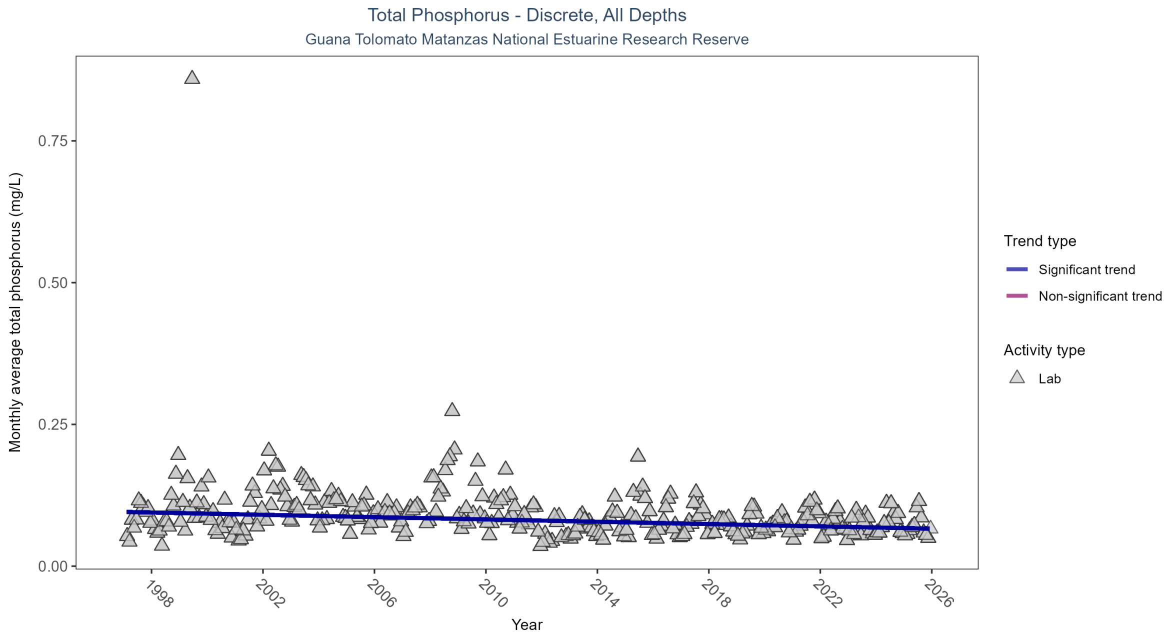

Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly decreasing trend	7339	29	1997 - 2025	0.07	-0.24616	0.09568	-0.00102	0

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

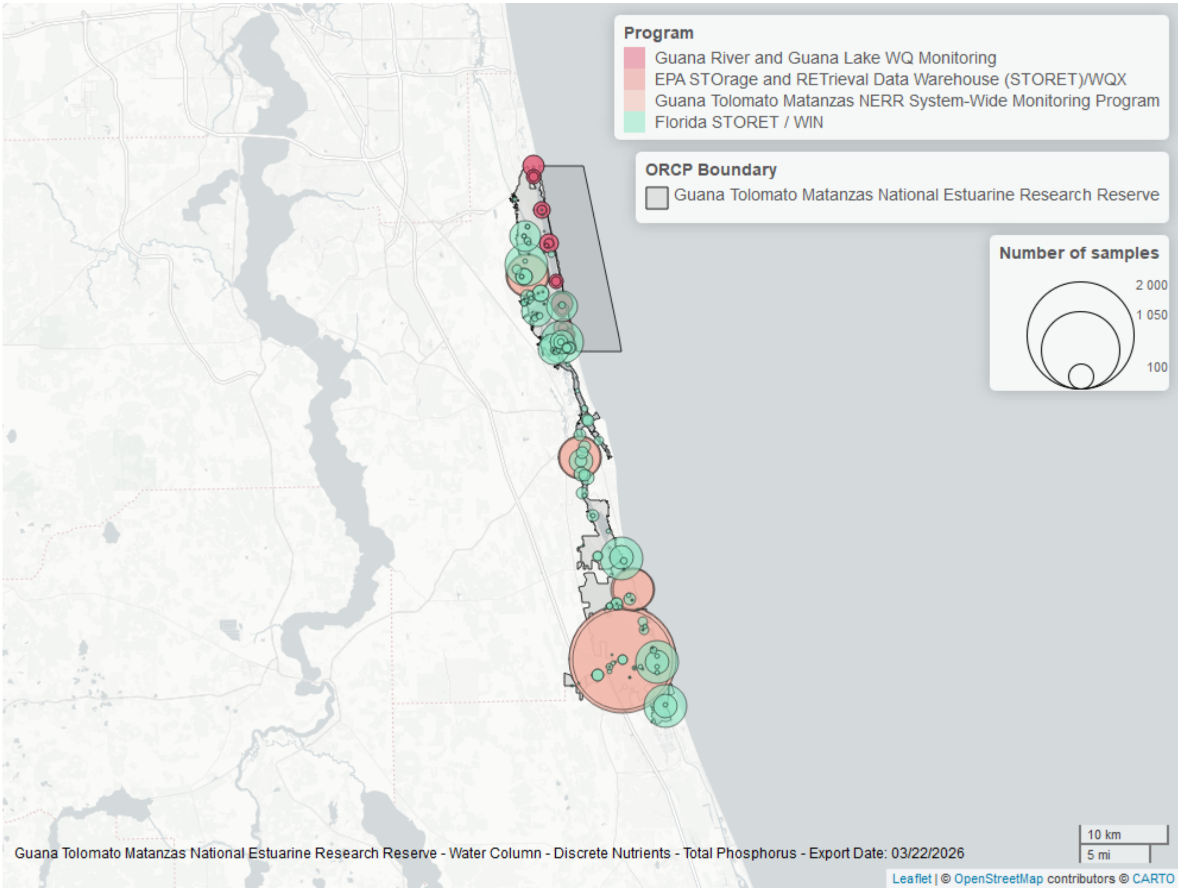

Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

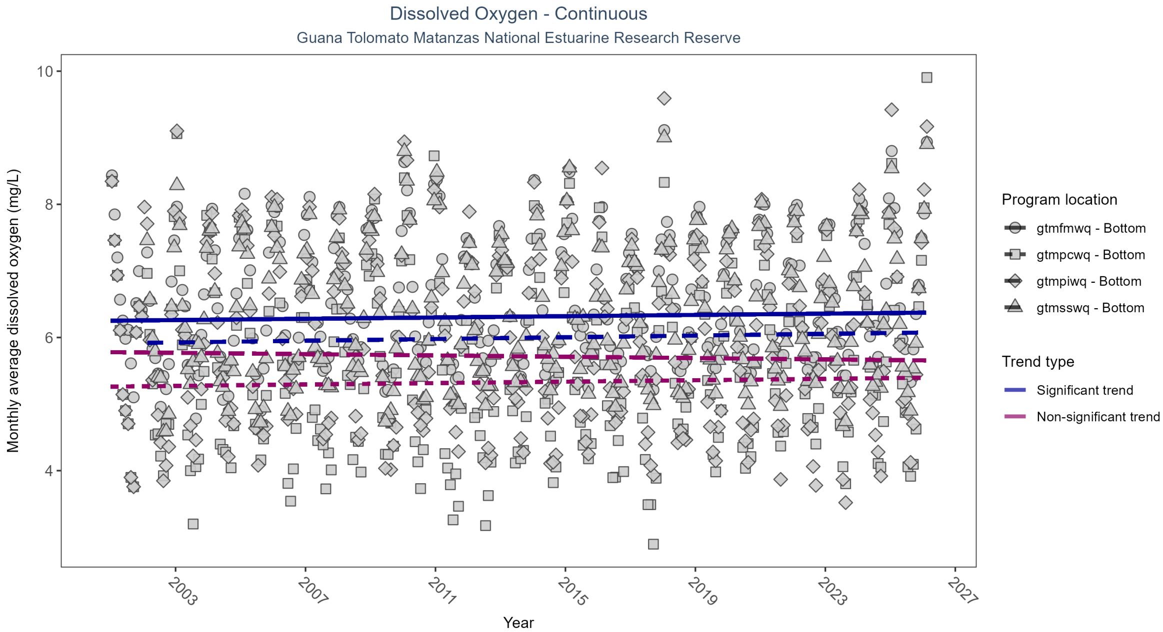

Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: Seasonal Kendall-Tau Results - Dissolved Oxygen

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	No significant trend	699401	26	2001 - 2026	5.9	-0.06	5.78	0.00	0.1298
gtmfmwq	Significantly increasing trend	718097	26	2001 - 2026	6.5	0.10	6.25	0.00	0.0153
gtmsswq	Significantly increasing trend	699026	25	2002 - 2026	6.3	0.10	5.92	0.01	0.0206
gtmpcwq	No significant trend	736088	26	2001 - 2026	5.5	0.05	5.26	0.01	0.1932

At two program locations, monthly average dissolved oxygen increased by less than 0.01 mg/L per year at one site and by 0.01 mg/L per year at the other. No detectable change in monthly average dissolved oxygen was observed at two locations.

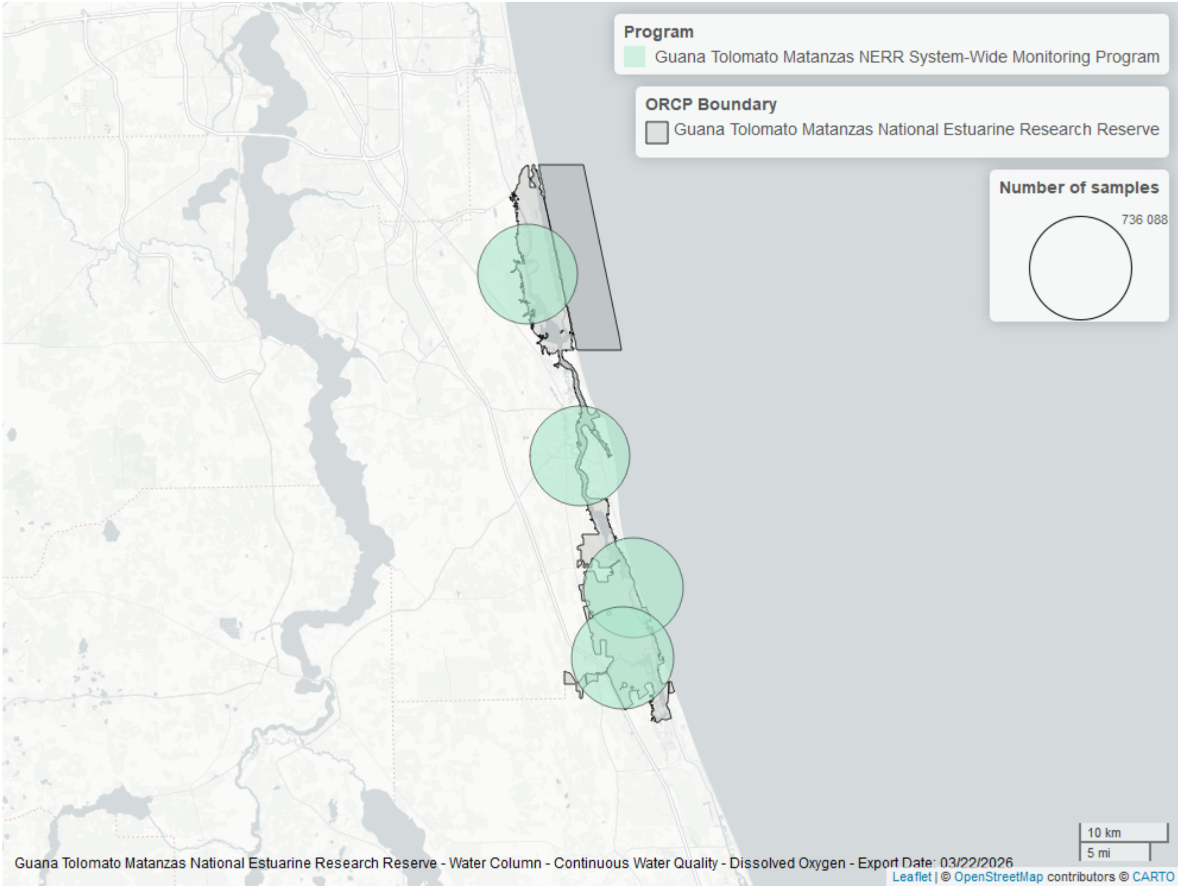

Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

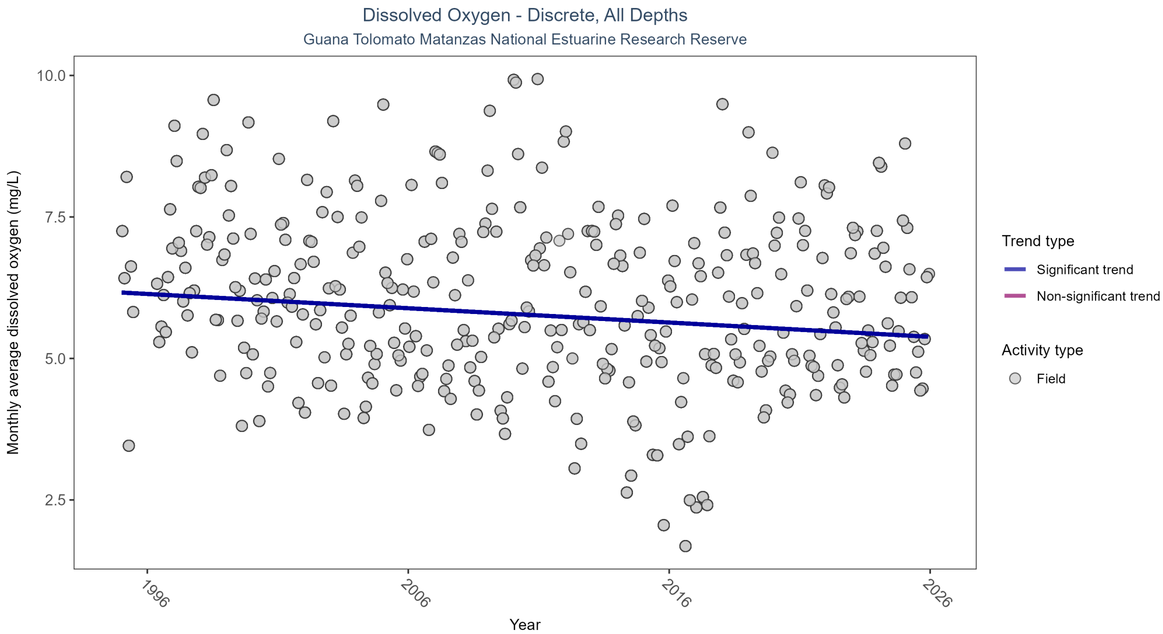

Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	21643	31	1995 - 2025	5.97	-0.16437	6.16464	-0.02524	0

Monthly average dissolved oxygen decreased by 0.03 mg/L per year.

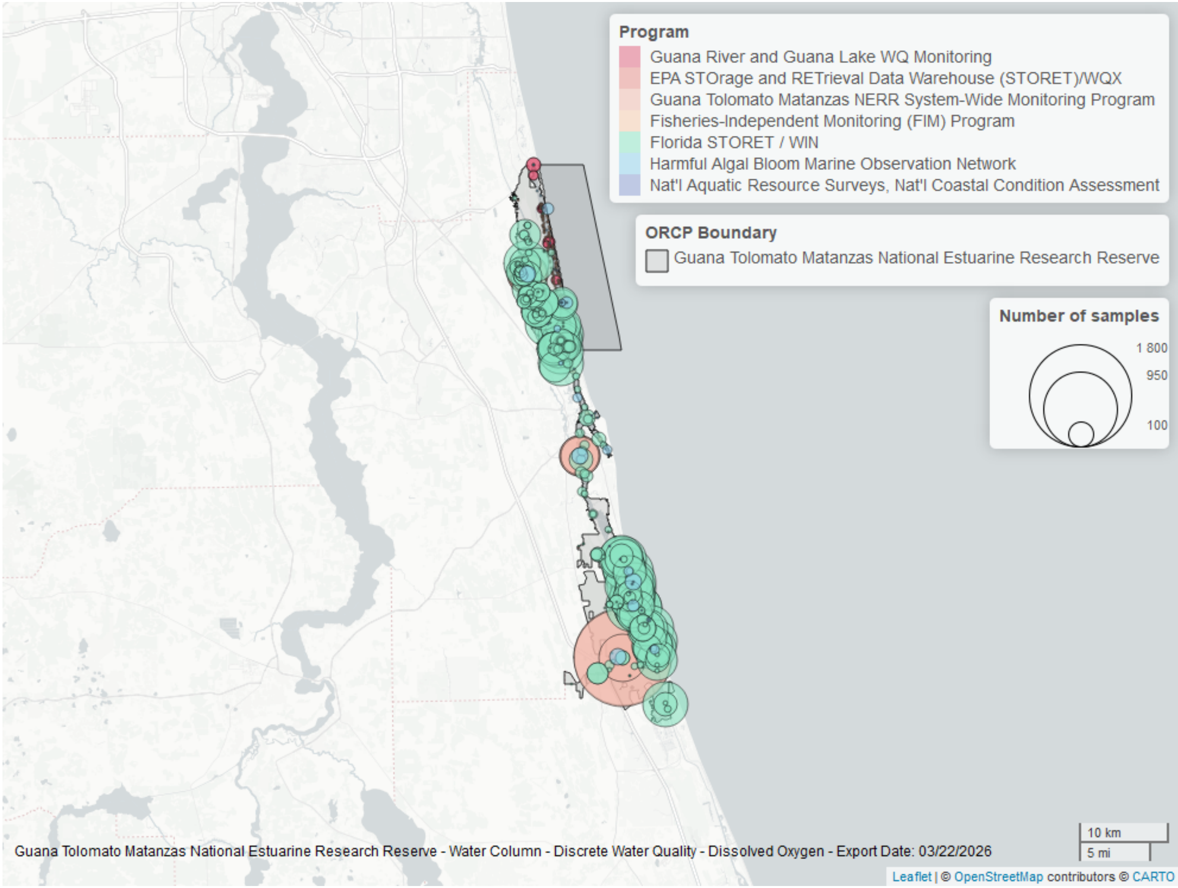

Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

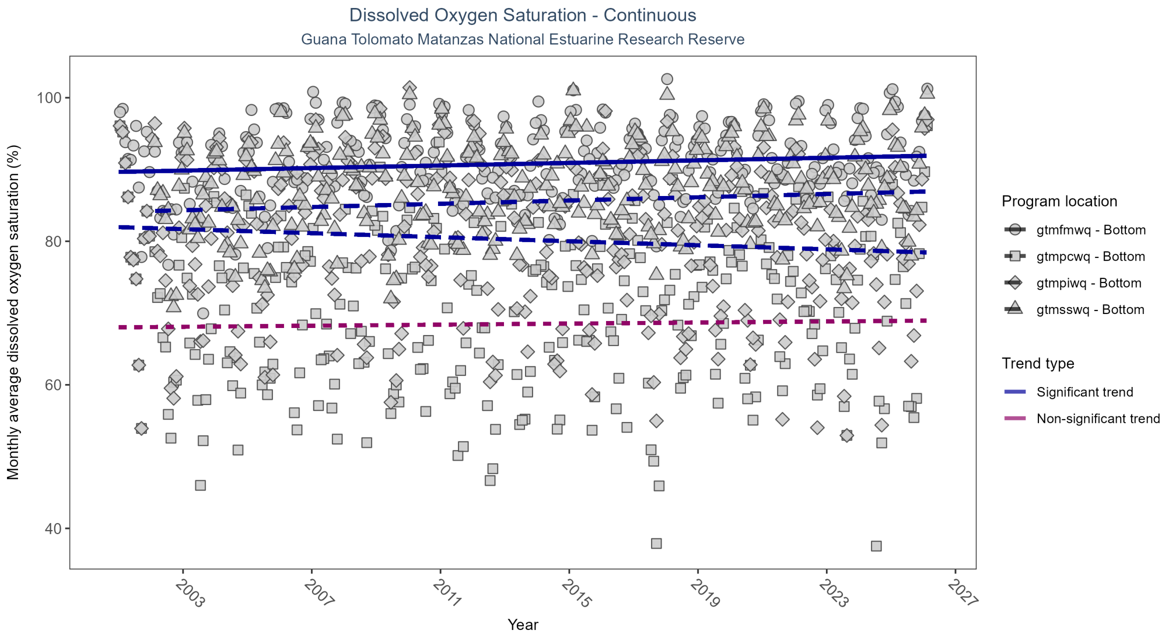

Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: Seasonal Kendall-Tau Results - Dissolved Oxygen Saturation

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Significantly decreasing trend	705219	26	2001 - 2026	82.2	-0.15	81.97	-0.14	3e-04
gtmfmwq	Significantly increasing trend	731140	26	2001 - 2026	92.7	0.16	89.66	0.09	1e-04
gtmsswq	Significantly increasing trend	704965	25	2002 - 2026	89.5	0.16	84.21	0.11	2e-04
gtmpcwq	No significant trend	737581	26	2001 - 2026	71.5	0.03	68.01	0.04	0.4939

At two program locations, monthly average dissolved oxygen saturation increased by 0.09% per year at one site and by 0.11% per year at the other. At one program location, monthly average dissolved oxygen saturation decreased by 0.14% per year. No detectable change in monthly average dissolved oxygen saturation was observed at one location.

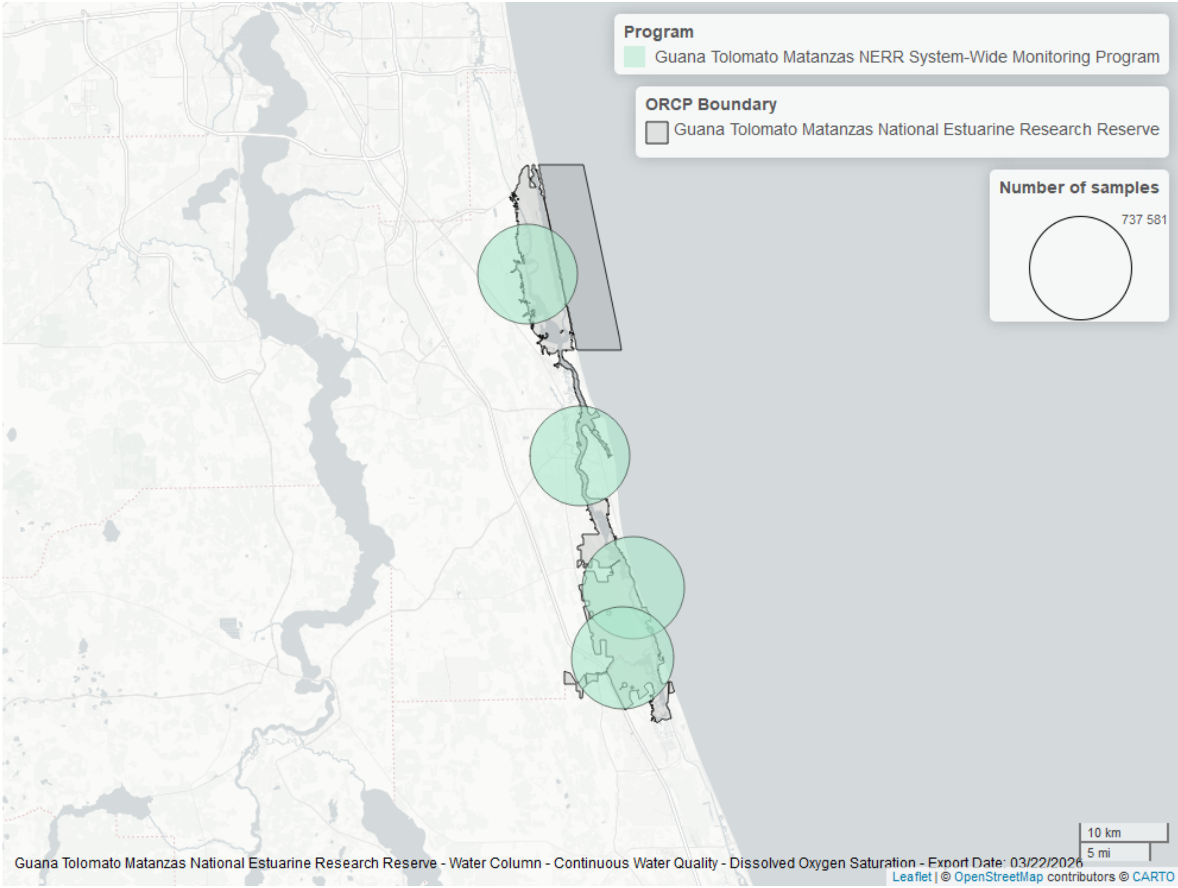

Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

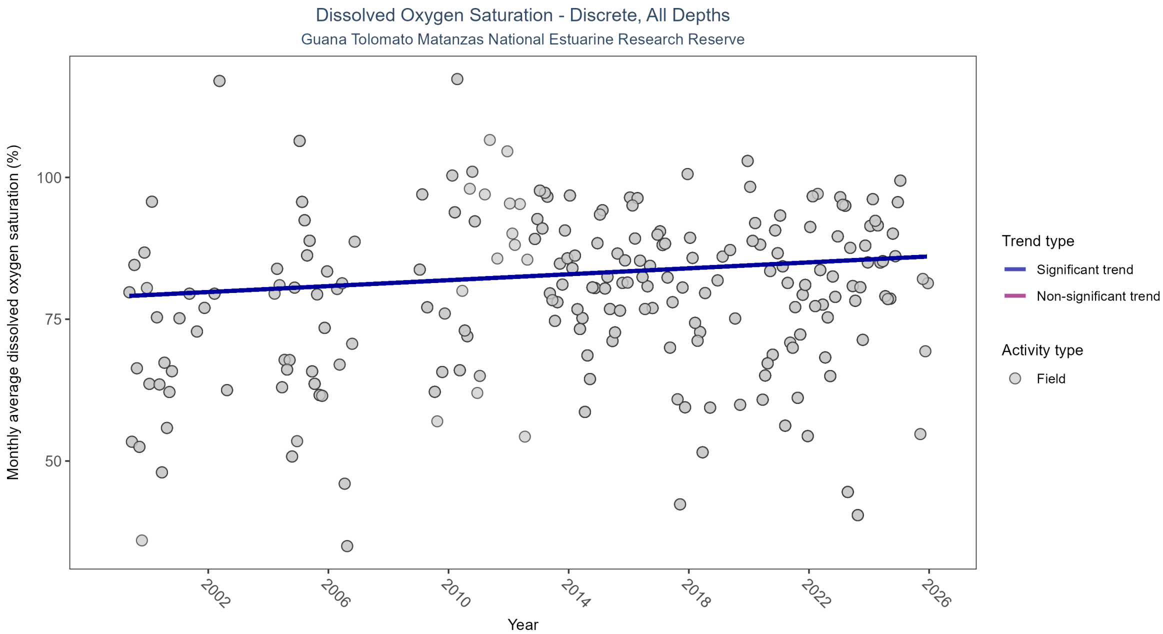

Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	2522	24	1999 - 2025	81.7	0.11881	79.00647	0.26134	0.0151

Monthly average dissolved oxygen saturation increased by 0.26% per year.

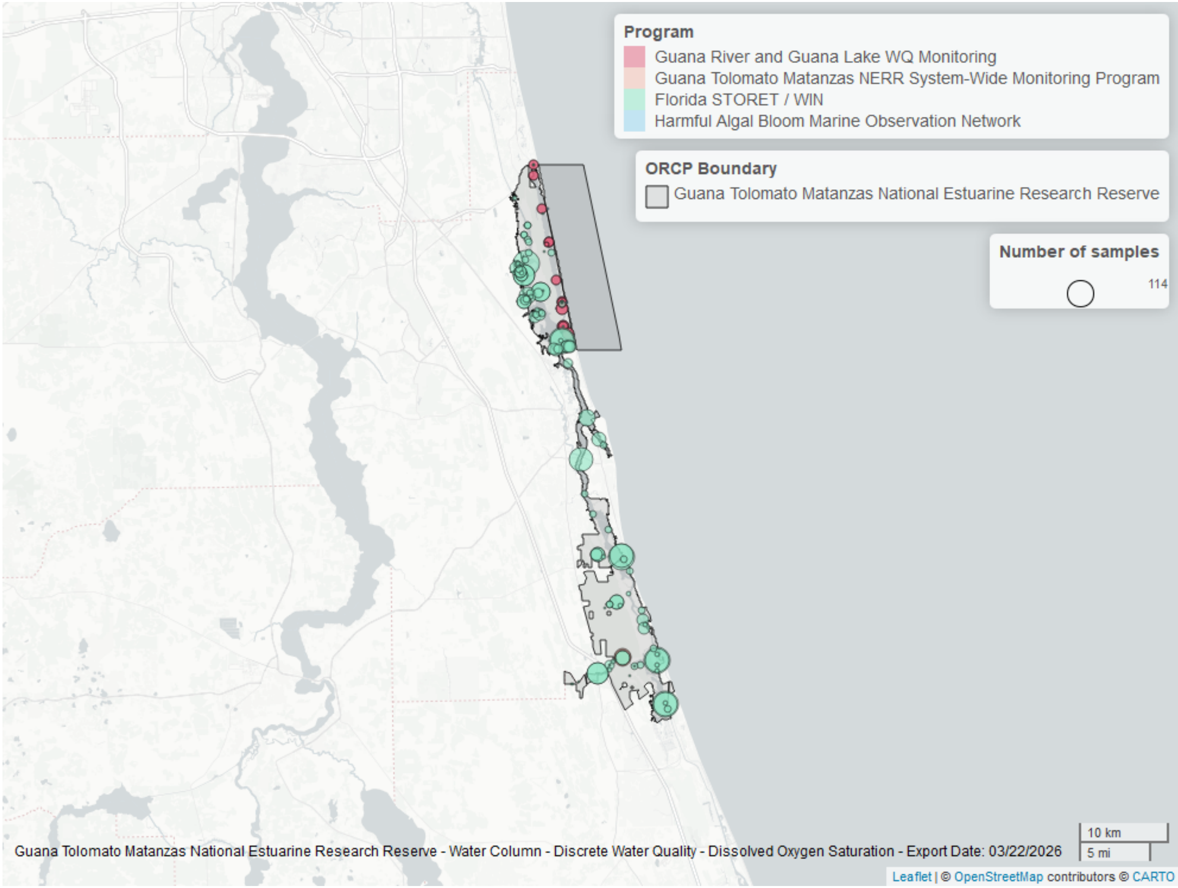

Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

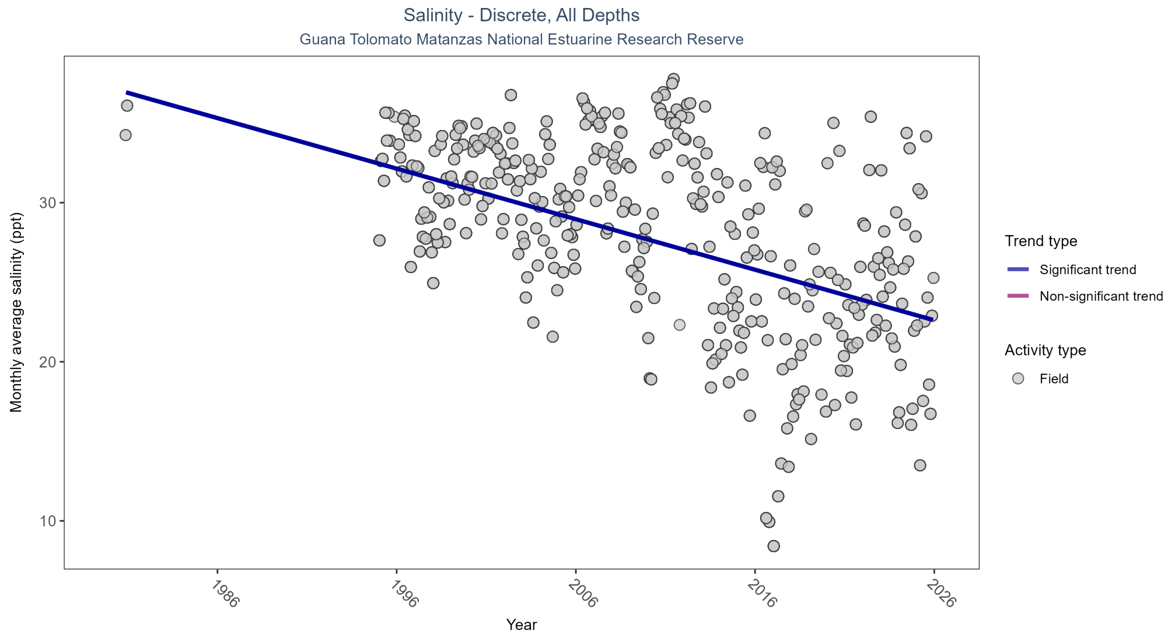

Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Salinity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Significantly decreasing trend	26181	32	1980 - 2025	31.7	-0.36753	37.22153	-0.31767	0

Monthly average salinity decreased by 0.32 ppt per year.

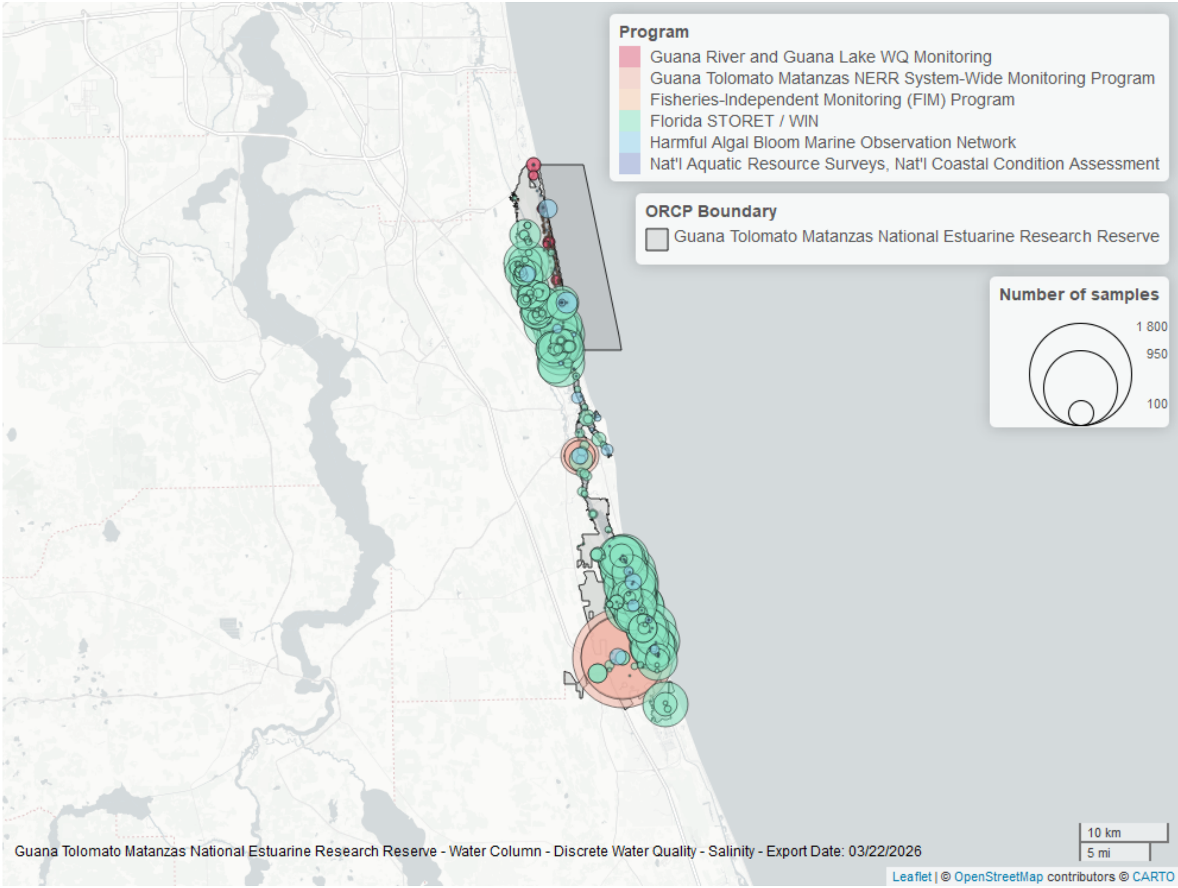

Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

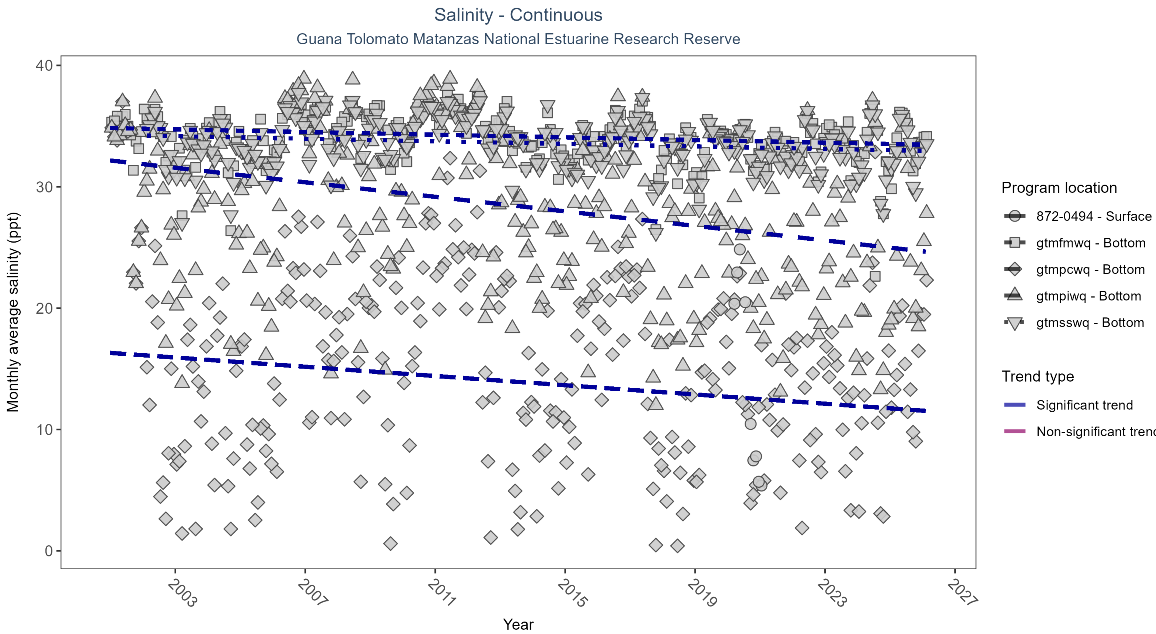

Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: Seasonal Kendall-Tau Results - Salinity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Significantly decreasing trend	703518	26	2001 - 2026	27.70	-0.26	32.16	-0.3	0
872-0494	Insufficient data to calculate trend	34918	2	2020 - 2021	8.99	-	-	-	-
gtmfmwq	Significantly decreasing trend	705987	26	2001 - 2026	34.40	-0.16	34.84	-0.05	1e-04
gtmsswq	Significantly decreasing trend	683666	25	2002 - 2026	33.90	-0.12	34.22	-0.05	0.004
gtmpcwq	Significantly decreasing trend	745868	26	2001 - 2026	16.60	-0.12	16.31	-0.19	0.0027

At four program locations, monthly average salinity decreased between 0.05 and 0.3 ppt per year. There was insufficient data to fit a model for one location.

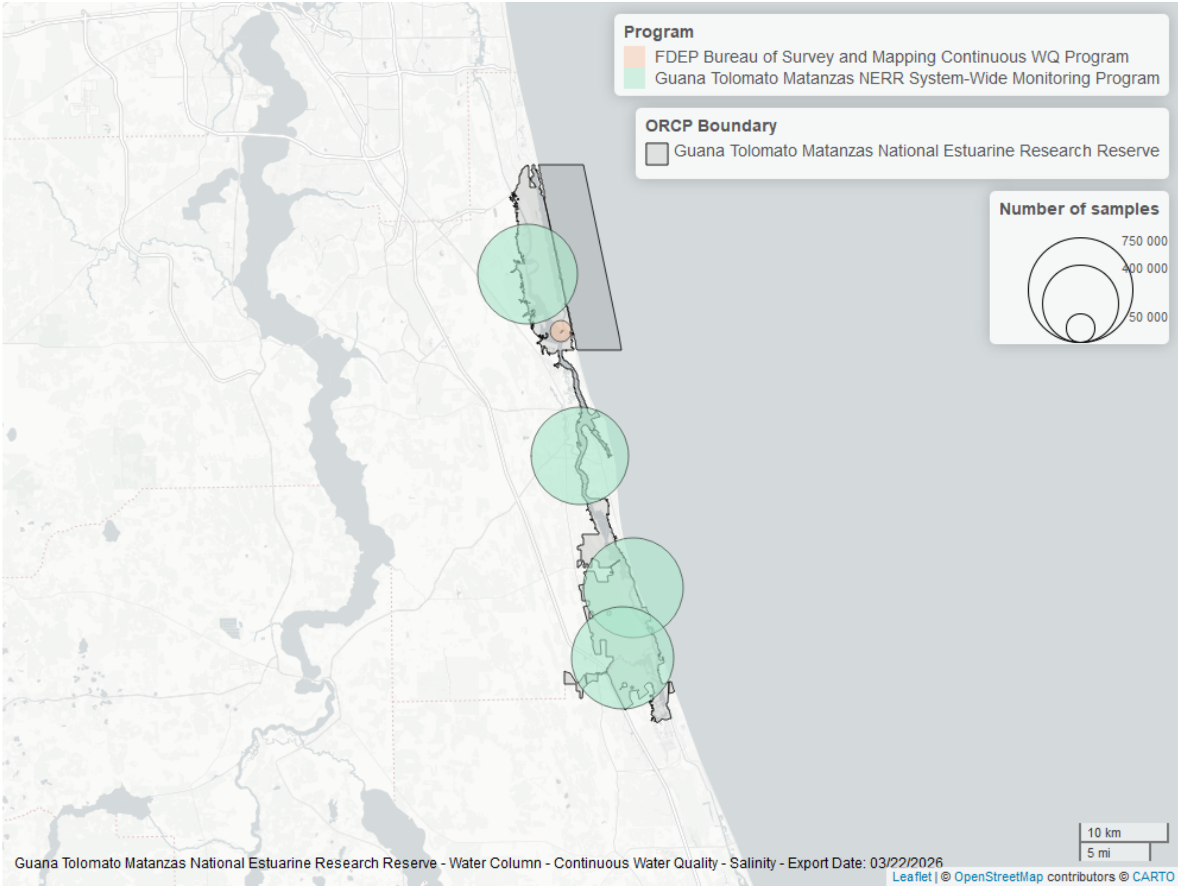

Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

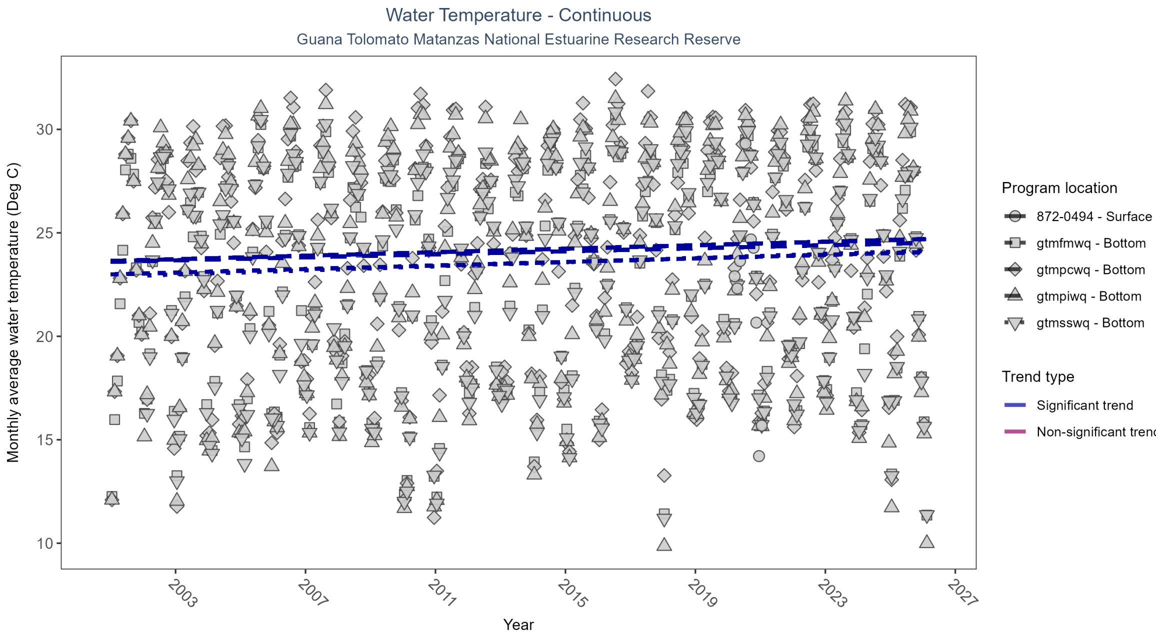

Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: Seasonal Kendall-Tau Results - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Significantly increasing trend	753811	26	2001 - 2026	24.20	0.18	23.6	0.04	0
872-0494	Insufficient data to calculate trend	35473	2	2020 - 2021	22.34	-	-	-	-
gtmfmwq	Significantly increasing trend	751023	26	2001 - 2026	23.70	0.21	22.99	0.04	0
gtmsswq	Significantly increasing trend	717305	25	2002 - 2026	23.80	0.22	22.94	0.05	0
gtmpcwq	Significantly increasing trend	754434	26	2001 - 2026	24.30	0.15	23.64	0.04	2e-04

At four program locations, monthly average water temperature increased between 0.04 and 0.05°C per year. There was insufficient data to fit a model for one location.

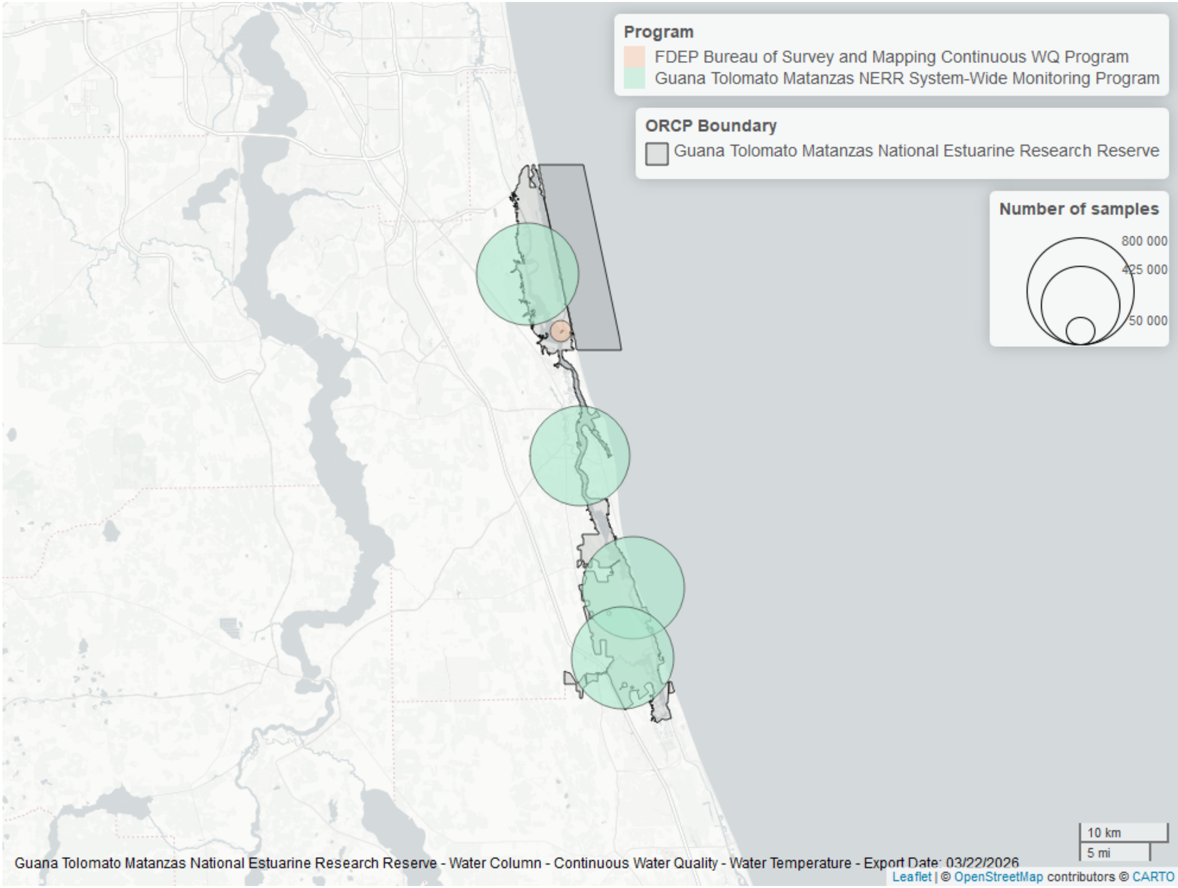

Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

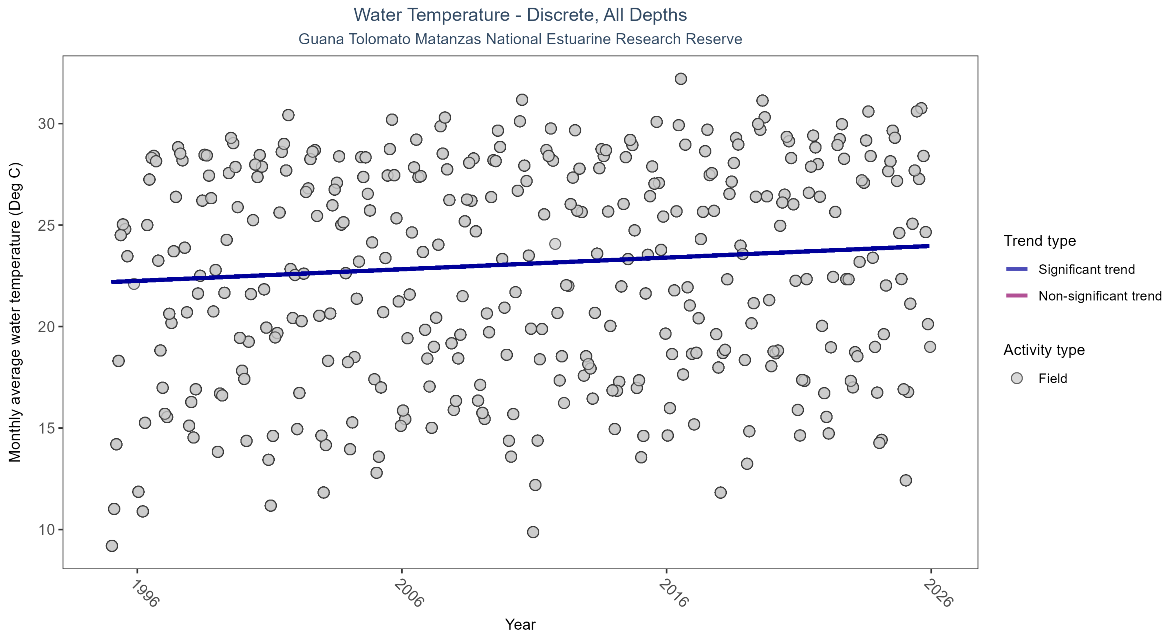

Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Results for - Water Temperature

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	24524	31	1995 - 2025	23.3	0.24139	22.19334	0.05734	0

Monthly average water temperature increased by 0.06°C per year.

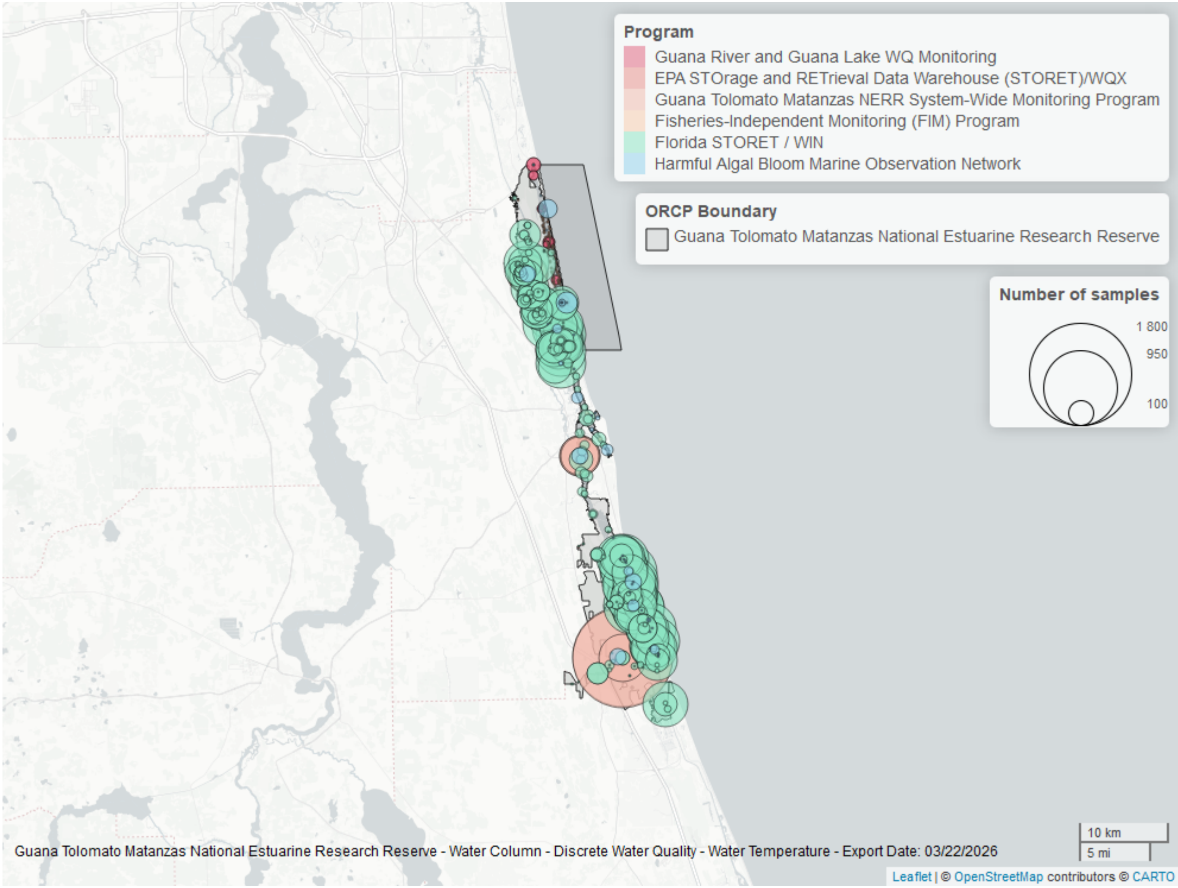

Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

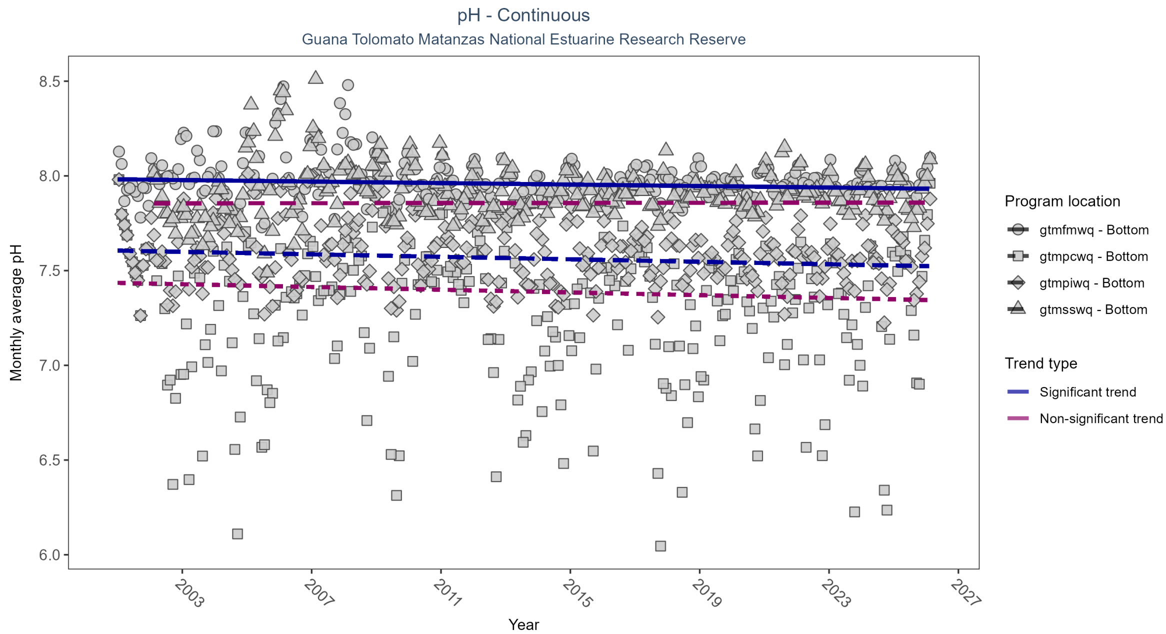

Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: Seasonal Kendall-Tau Results - pH

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Significantly decreasing trend	696449	26	2001 - 2026	7.6	-0.17	7.61	0	0
gtmfmwq	Significantly decreasing trend	702168	26	2001 - 2026	8.0	-0.14	7.98	0	6e-04
gtmsswq	No significant trend	674734	25	2002 - 2026	7.9	0.01	7.86	0	0.8494
gtmpcwq	No significant trend	730315	26	2001 - 2026	7.4	-0.08	7.44	0	0.0568

At two program locations, monthly average pH decreased by 0.00 pH units per year. No detectable change in monthly average pH was observed at two locations.

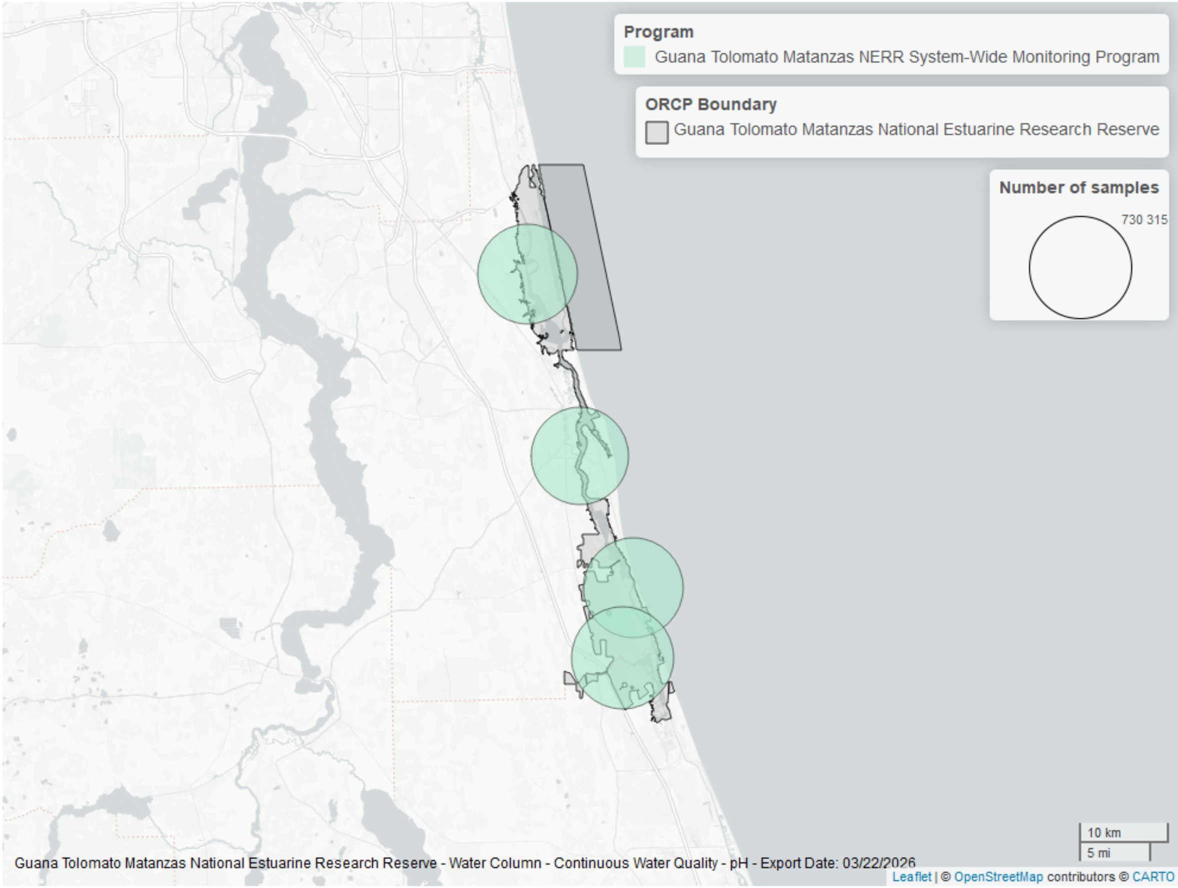

Figure 22: Map showing location of ph continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

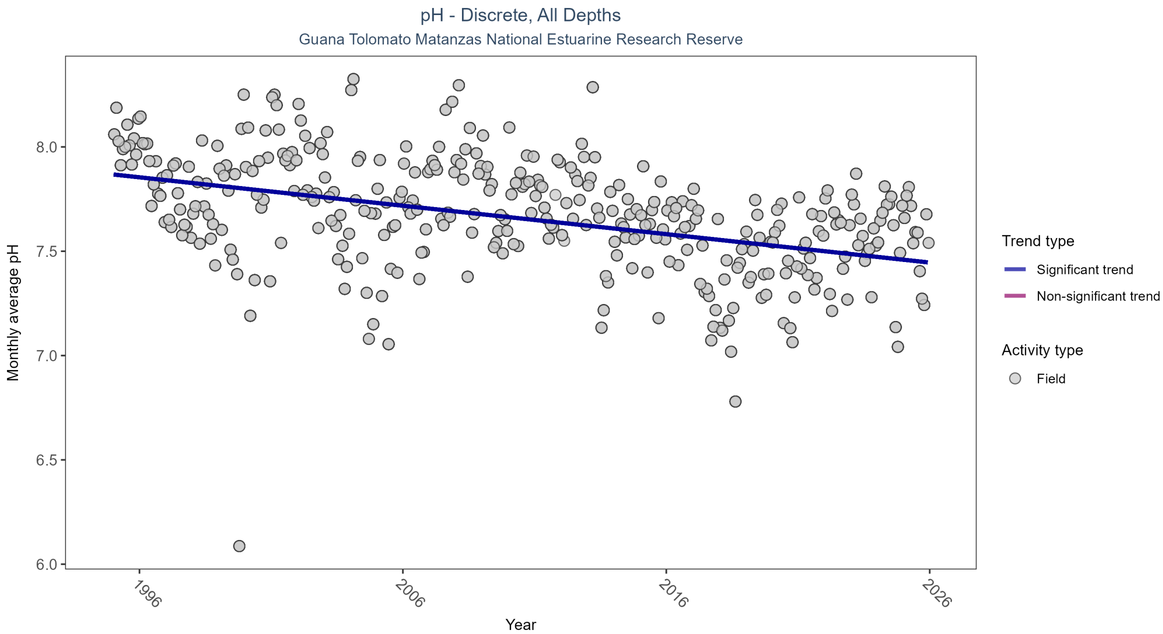

Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Results for - pH

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	17952	31	1995 - 2025	7.8	-0.37495	7.86792	-0.01362	0

Monthly average pH decreased by 0.01 pH units per year.

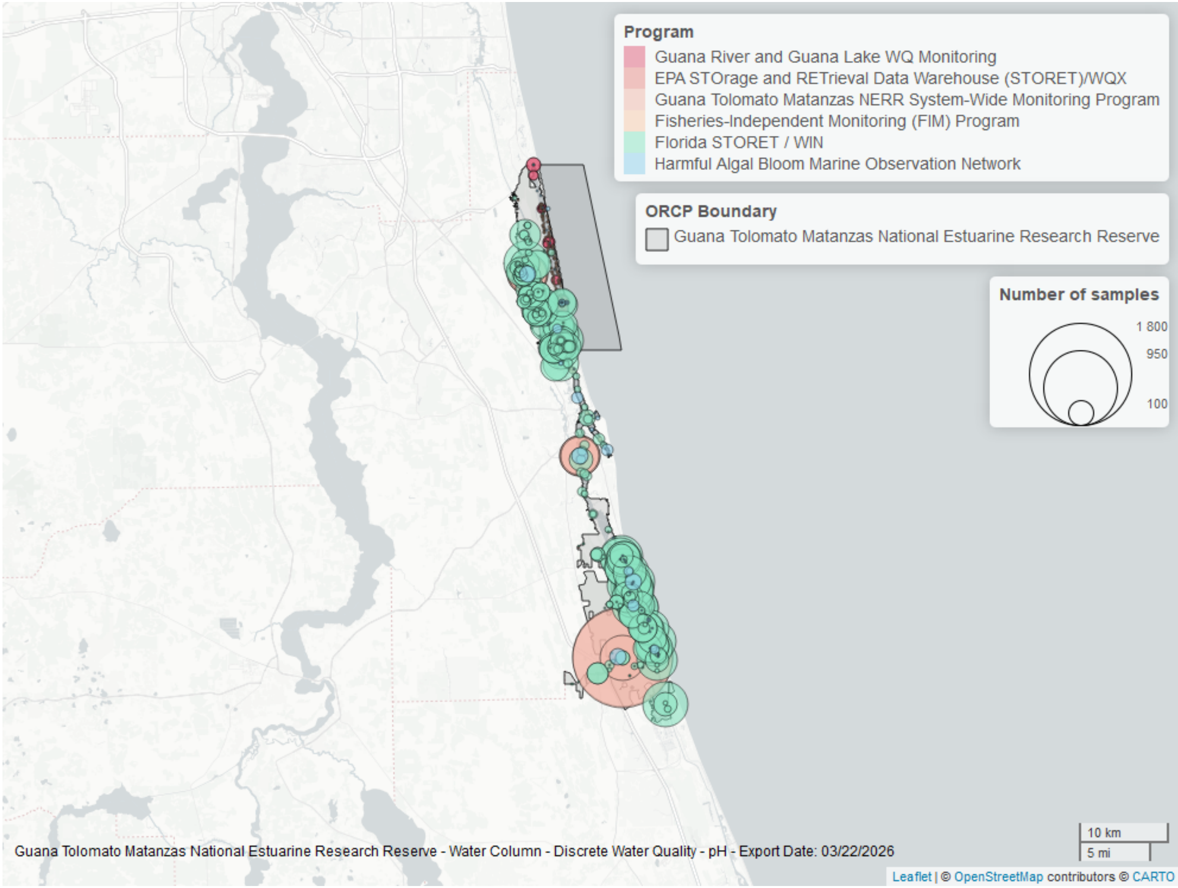

Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

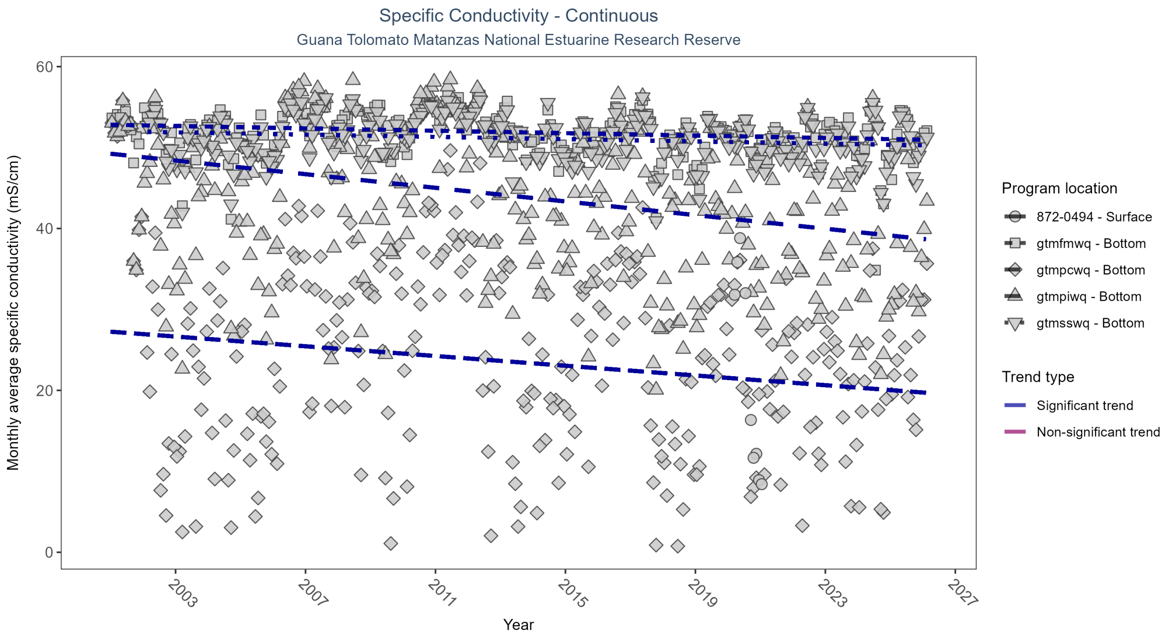

Table 13: Seasonal Kendall-Tau Results - Specific Conductivity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Significantly decreasing trend	703518	26	2001 - 2026	43.06	-0.27	49.24	-0.42	0
872-0494	Insufficient data to calculate trend	34894	2	2020 - 2021	14.10	-	-	-	-
gtmfmwq	Significantly decreasing trend	705986	26	2001 - 2026	52.22	-0.15	52.81	-0.07	2e-04
gtmsswq	Significantly decreasing trend	683665	25	2002 - 2026	51.54	-0.12	51.95	-0.07	0.0038
gtmpcwq	Significantly decreasing trend	746635	26	2001 - 2026	27.04	-0.13	27.25	-0.3	0.002

At four program locations, monthly average specific conductivity decreased between 0.07 and 0.42 ms/cm per year. There was insufficient data to fit a model for one location.

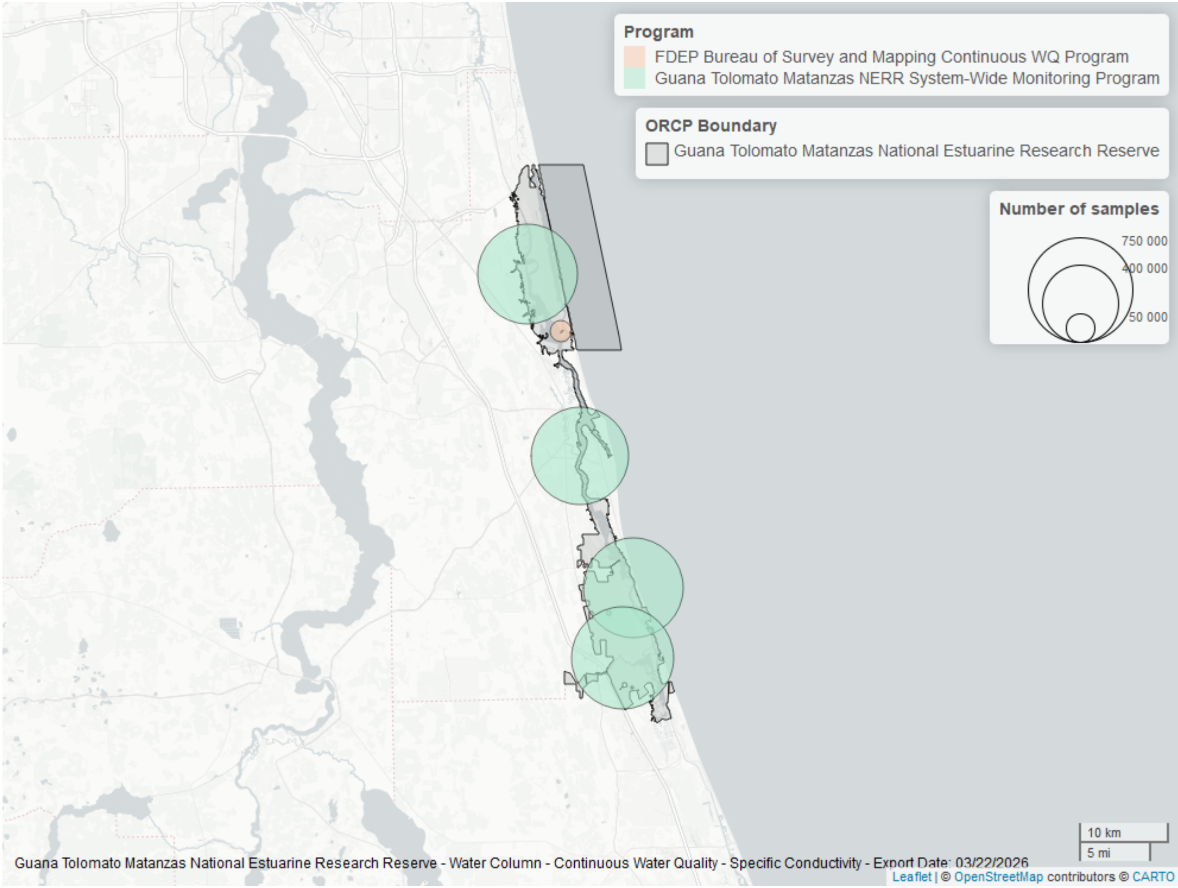

Figure 25: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

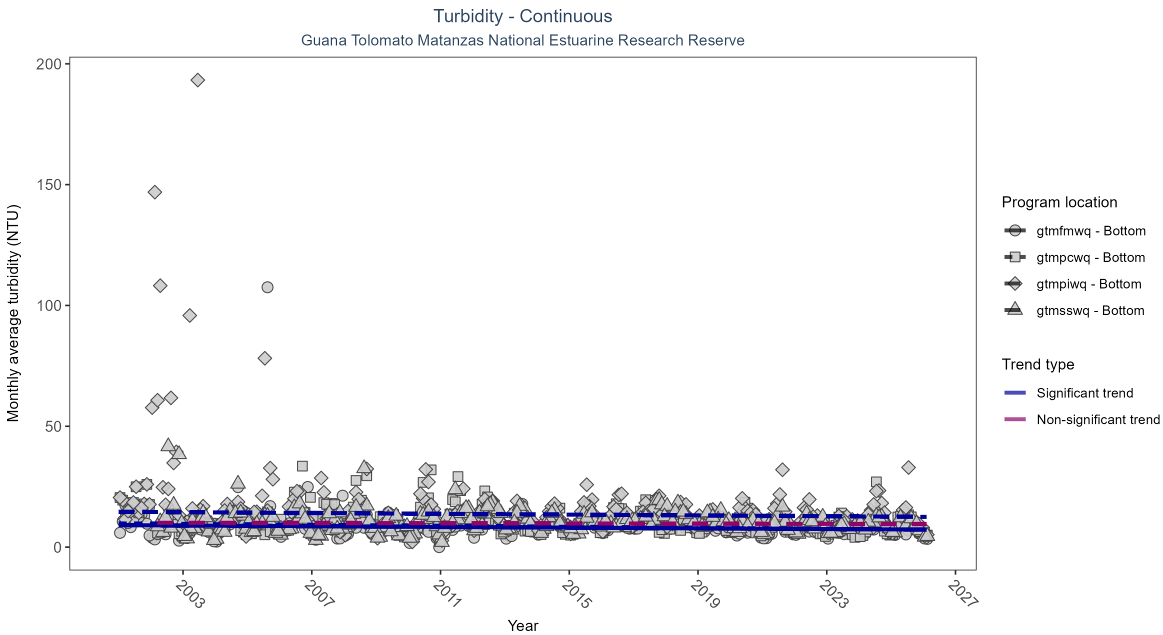

Figure 26: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: Seasonal Kendall-Tau Results - Turbidity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Significantly decreasing trend	679520	26	2001 - 2026	10	-0.12	14.61	-0.08	0.0048
gtmfmwq	Significantly decreasing trend	717246	26	2001 - 2026	7	-0.14	9.08	-0.07	0.001
gtmsswq	No significant trend	670272	25	2002 - 2026	9	-0.04	10.11	-0.02	0.3797
gtmpcwq	Significantly decreasing trend	721298	26	2001 - 2026	9	-0.17	9.80	-0.10	0

At three program locations, monthly average turbidity decreased between 0.07 and 0.1 NTU per year. No detectable change in monthly average turbidity was observed at one location.

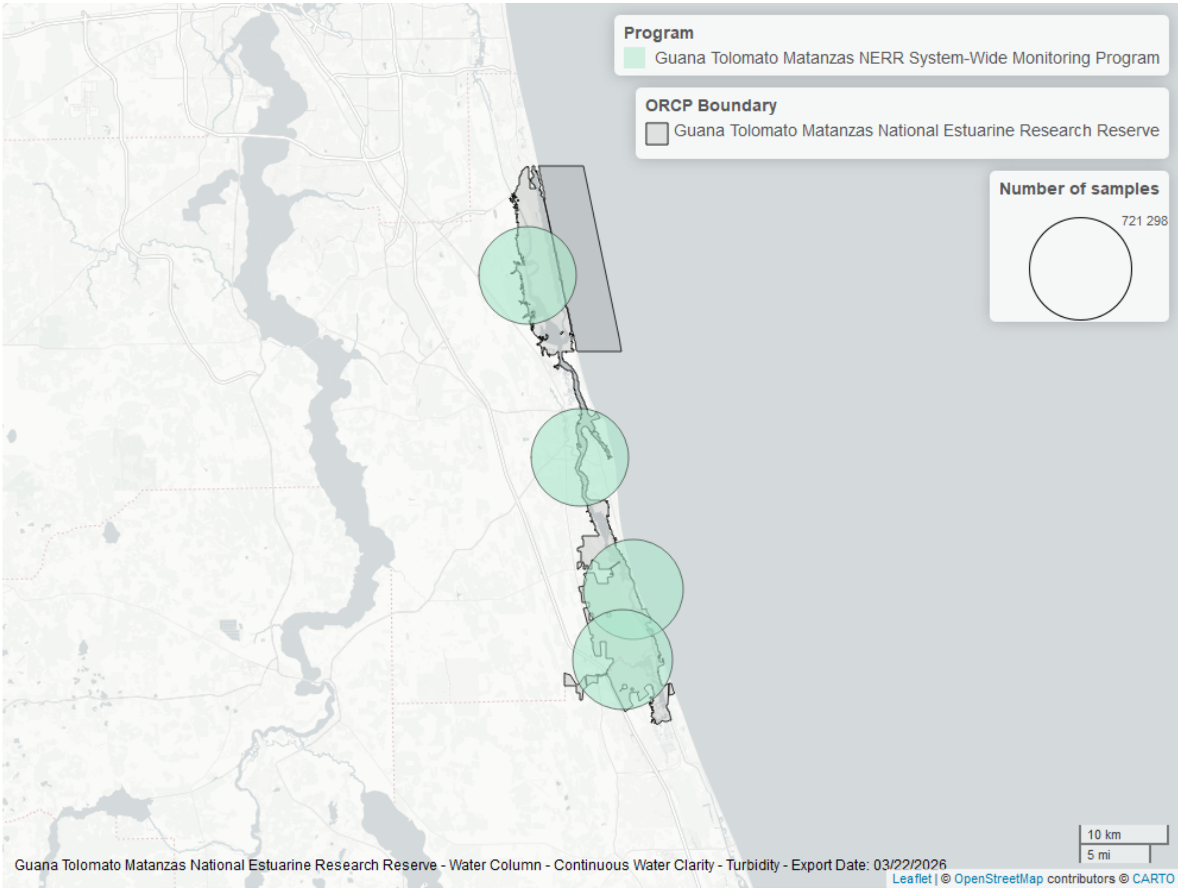

Figure 27: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

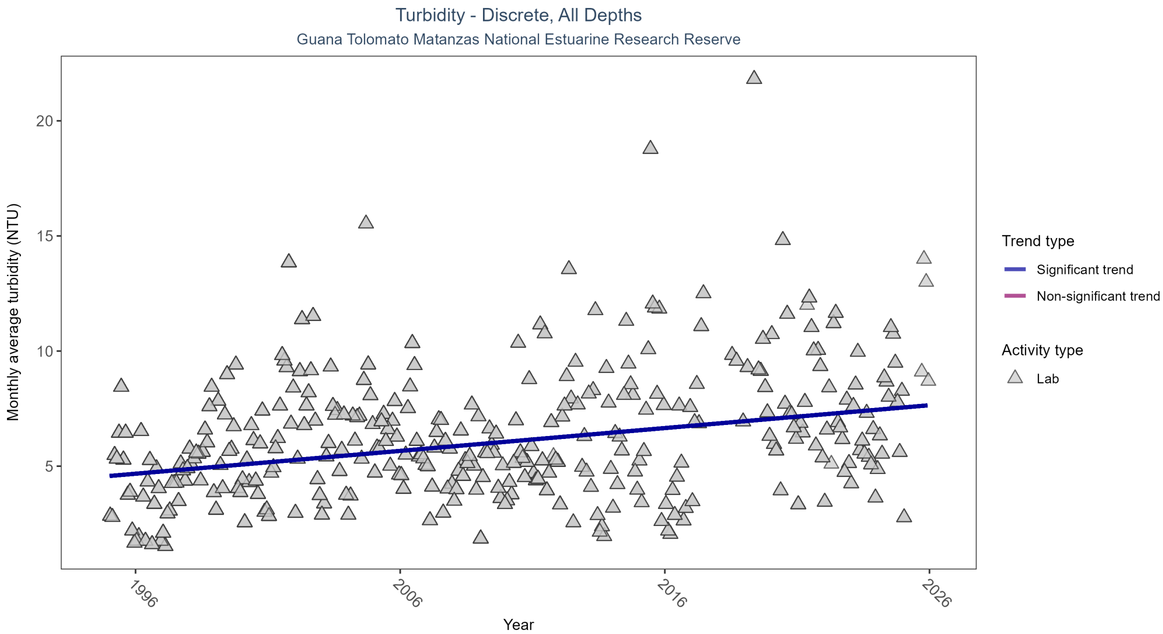

Figure 28: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: Seasonal Kendall-Tau Results for - Turbidity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	14885	31	1995 - 2025	4.5	0.26559	4.57195	0.09917	0

Monthly average turbidity increased by 0.1 NTU per year, indicating a decrease in water clarity.

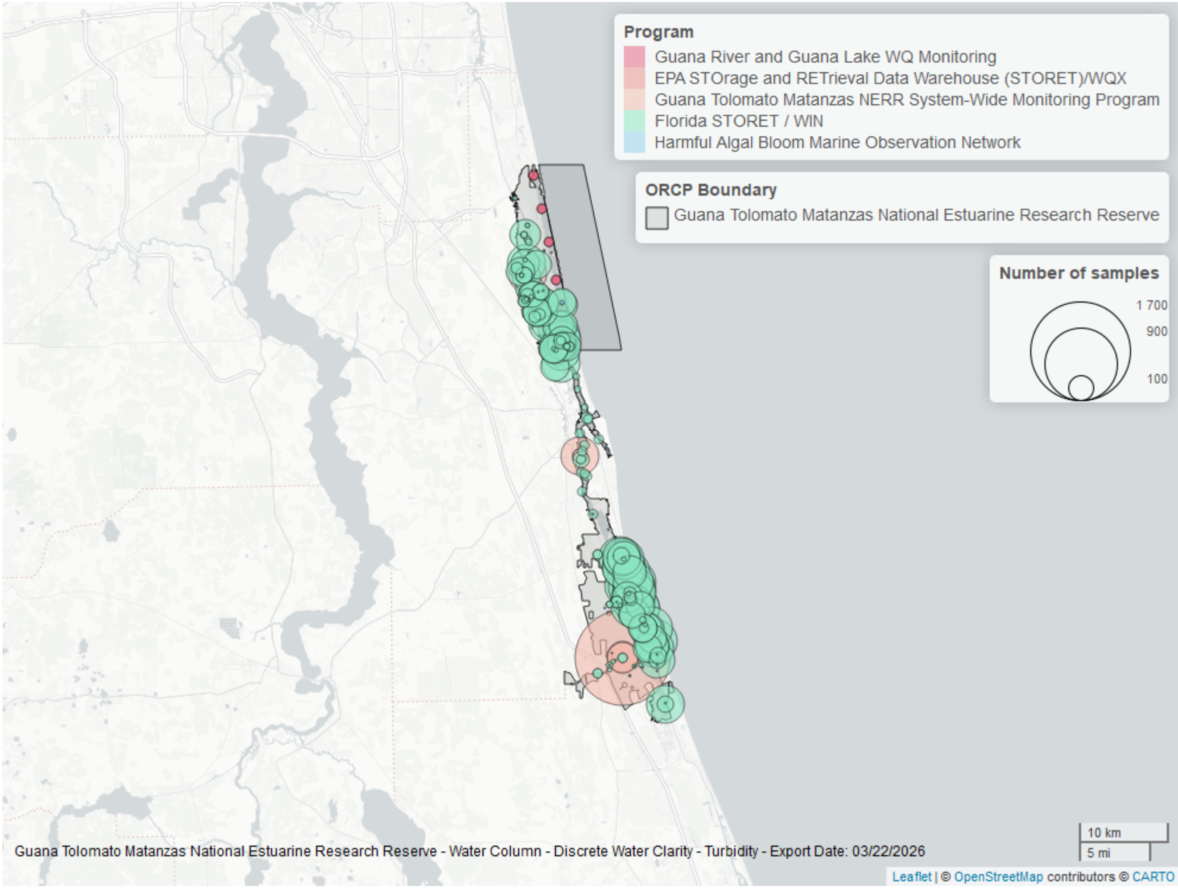

Figure 29: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

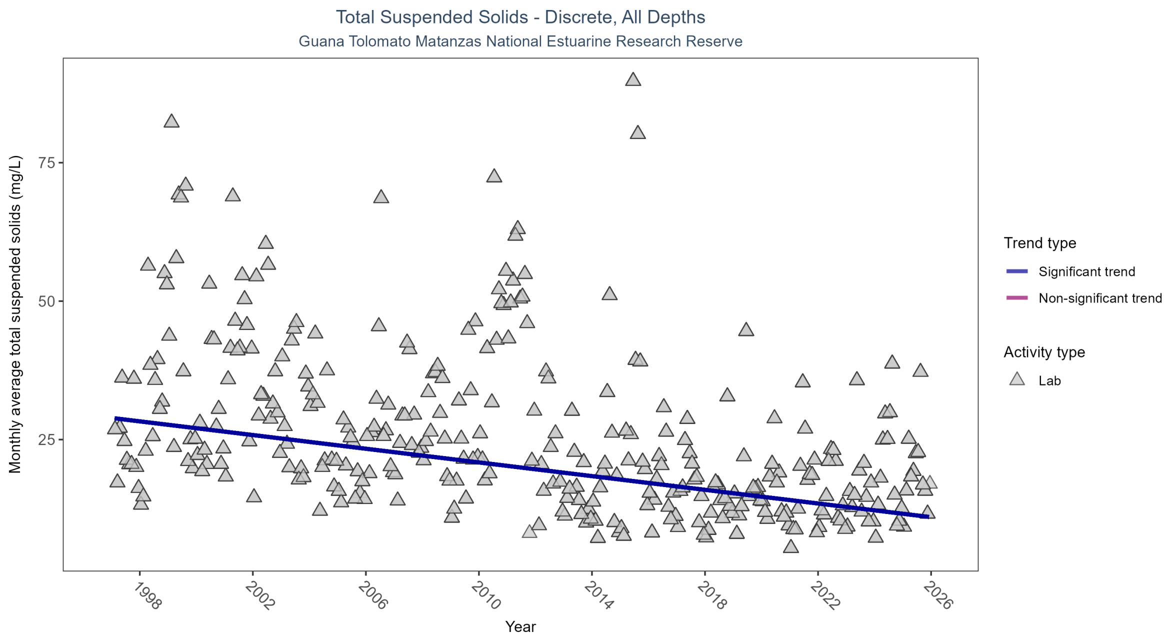

Figure 30: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Seasonal Kendall-Tau Results for - Total Suspended Solids

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly decreasing trend	5276	29	1997 - 2025	17	-0.38129	28.90548	-0.6179	0

Monthly average total suspended solids decreased by 0.62 mg/L per year, indicating an increase in water clarity.

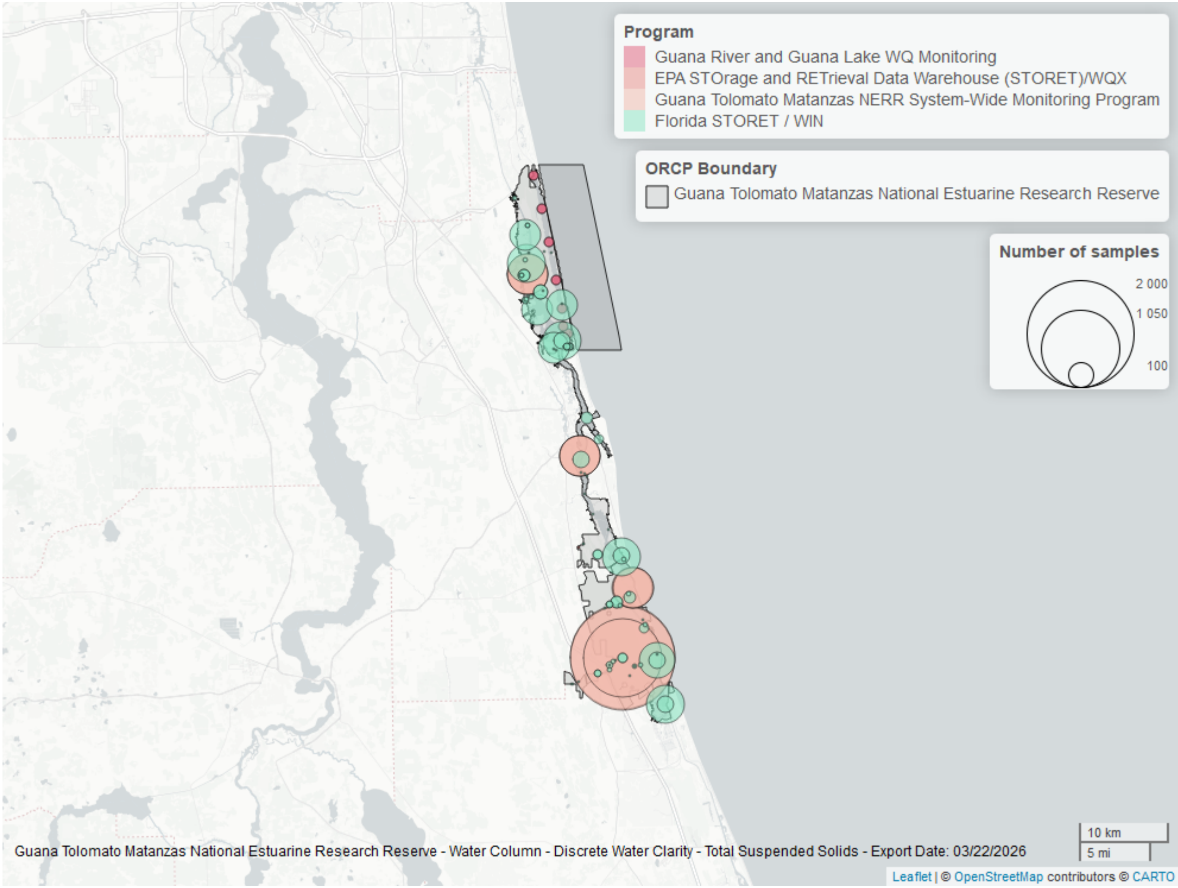

Figure 31: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

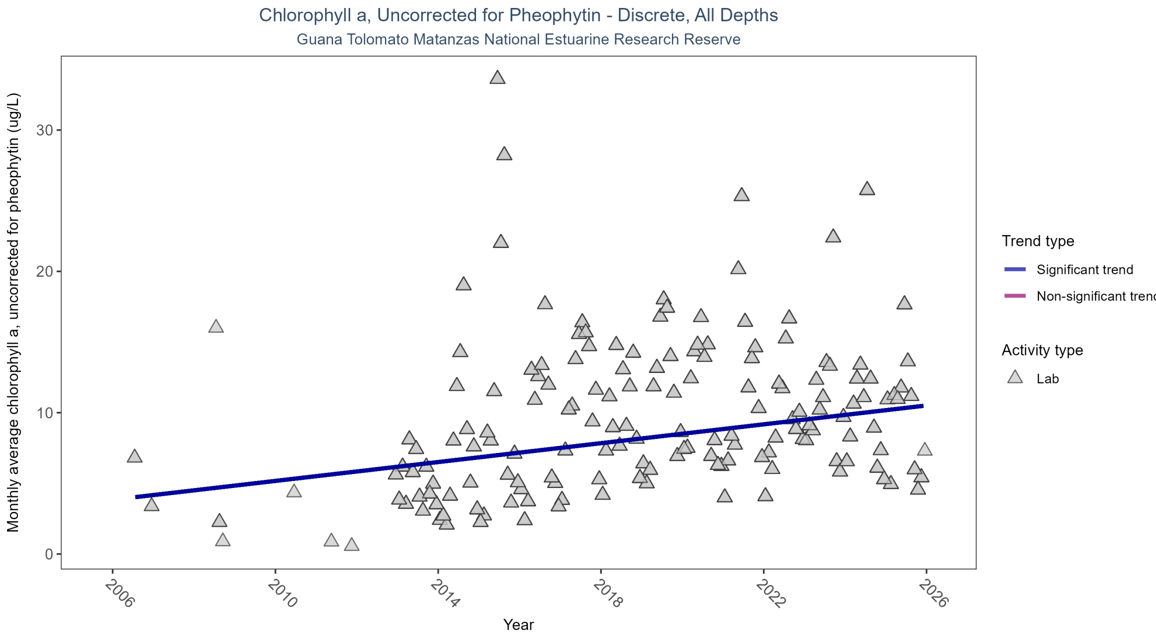

Figure 32: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	3970	18	2006 - 2025	6.3	0.28882	3.83917	0.33361	0

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.33 µg/L per year, indicating a decrease in water clarity.

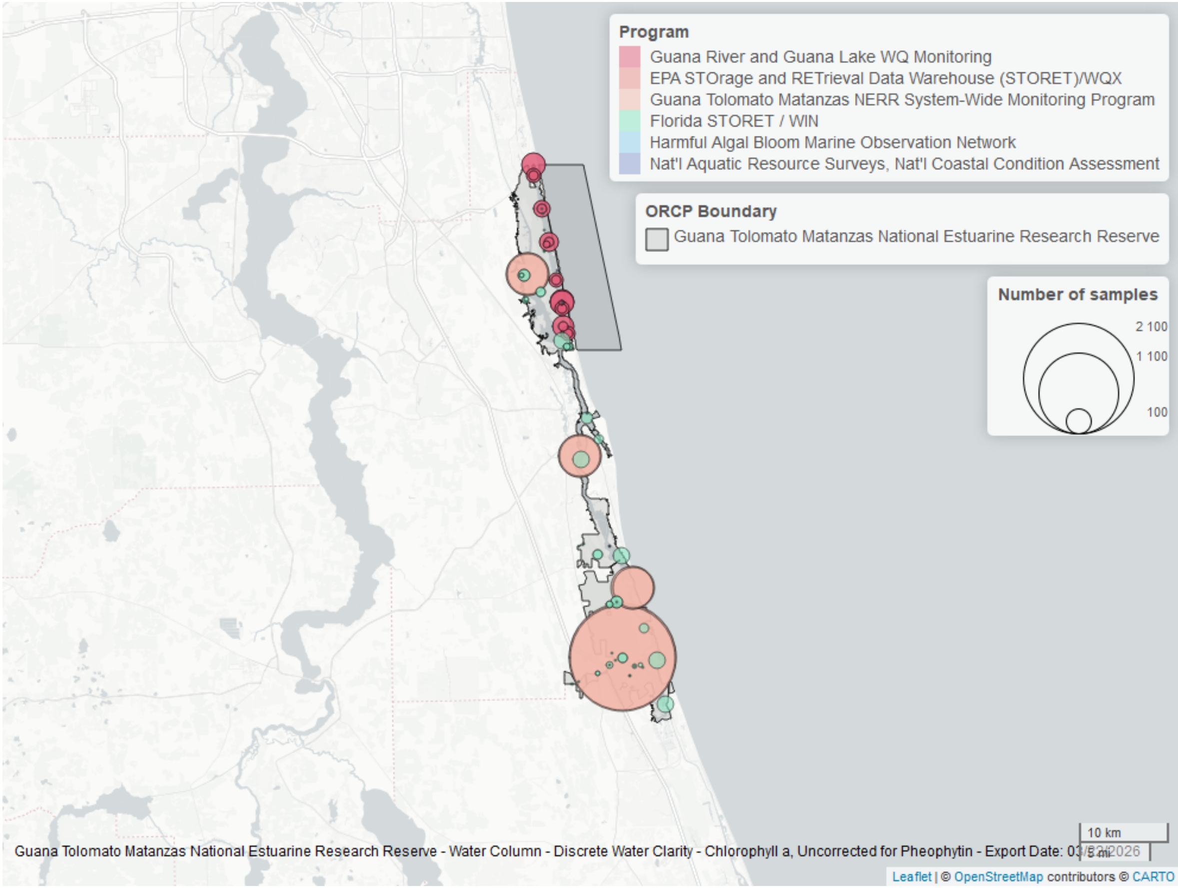

Figure 33: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

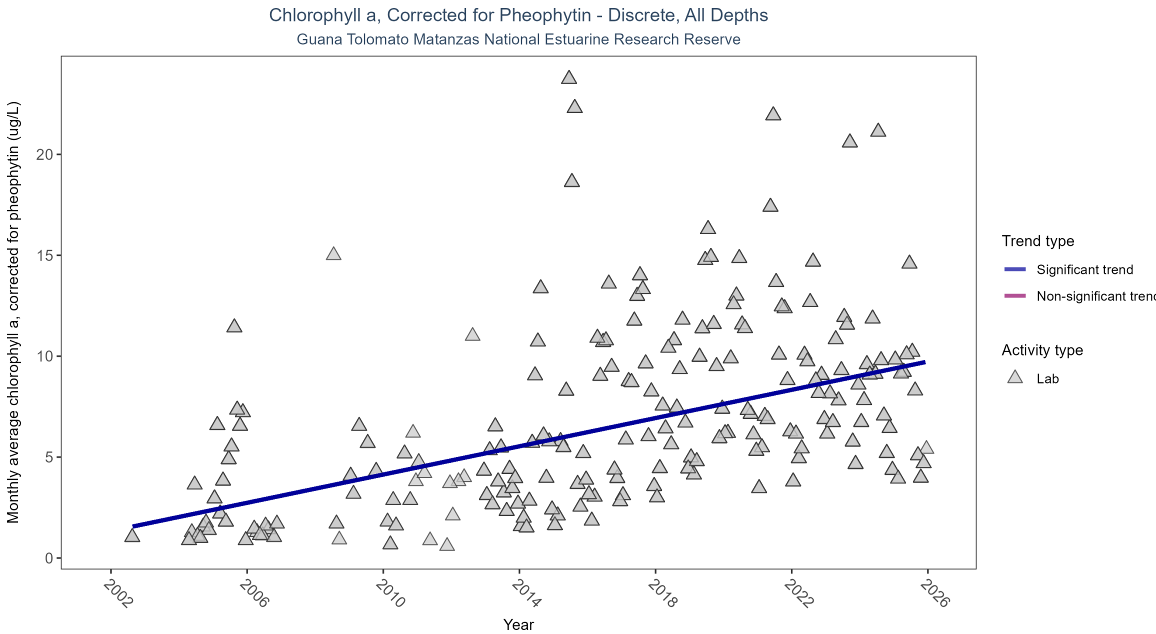

Figure 34: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	4349	22	2002 - 2025	4.7	0.43916	1.33651	0.34993	0

Monthly average chlorophyll a, corrected for pheophytin, increased by 0.35 µg/L per year, indicating a decrease in water clarity.

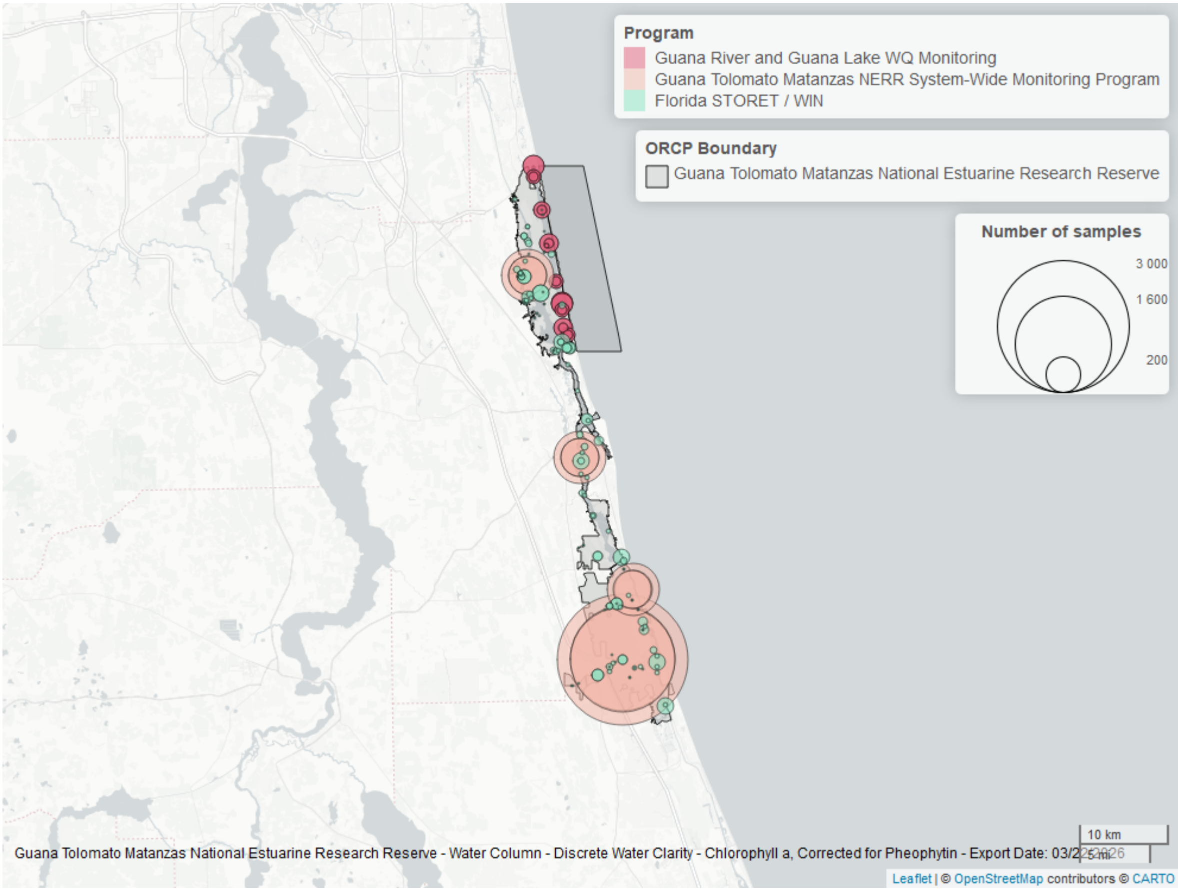

Figure 35: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

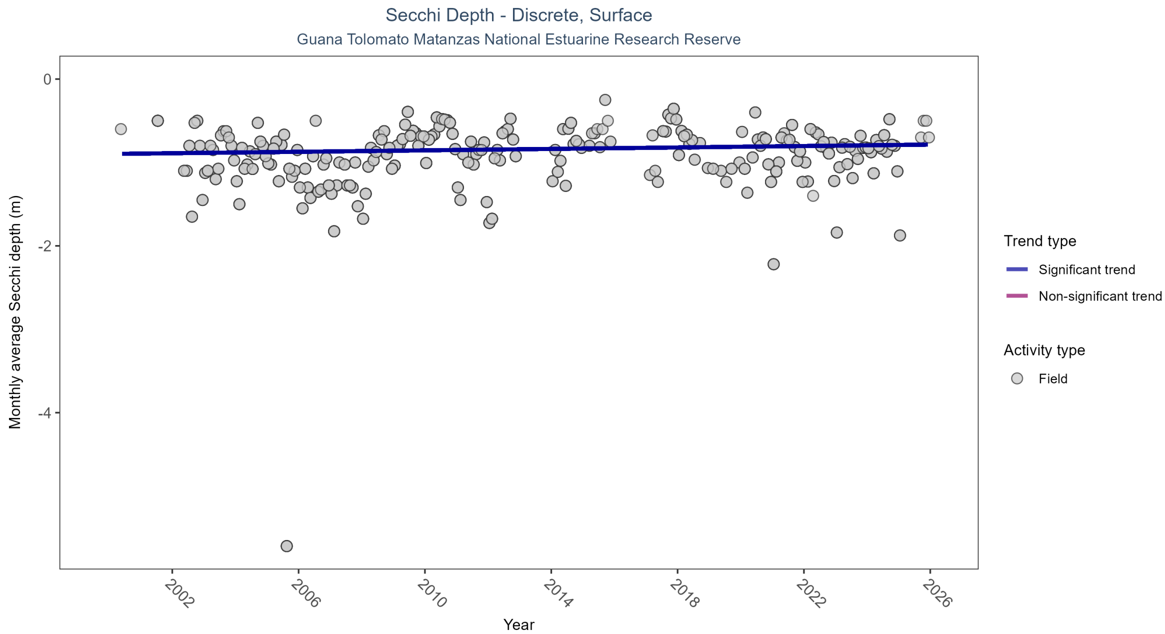

Figure 36: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Seasonal Kendall-Tau Results for - Secchi Depth

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	1867	24	2000 - 2025	-0.8	0.10701	-0.89795	0.00432	0.0316

Secchi depth showed no detectable trend between 1999 and 2025.

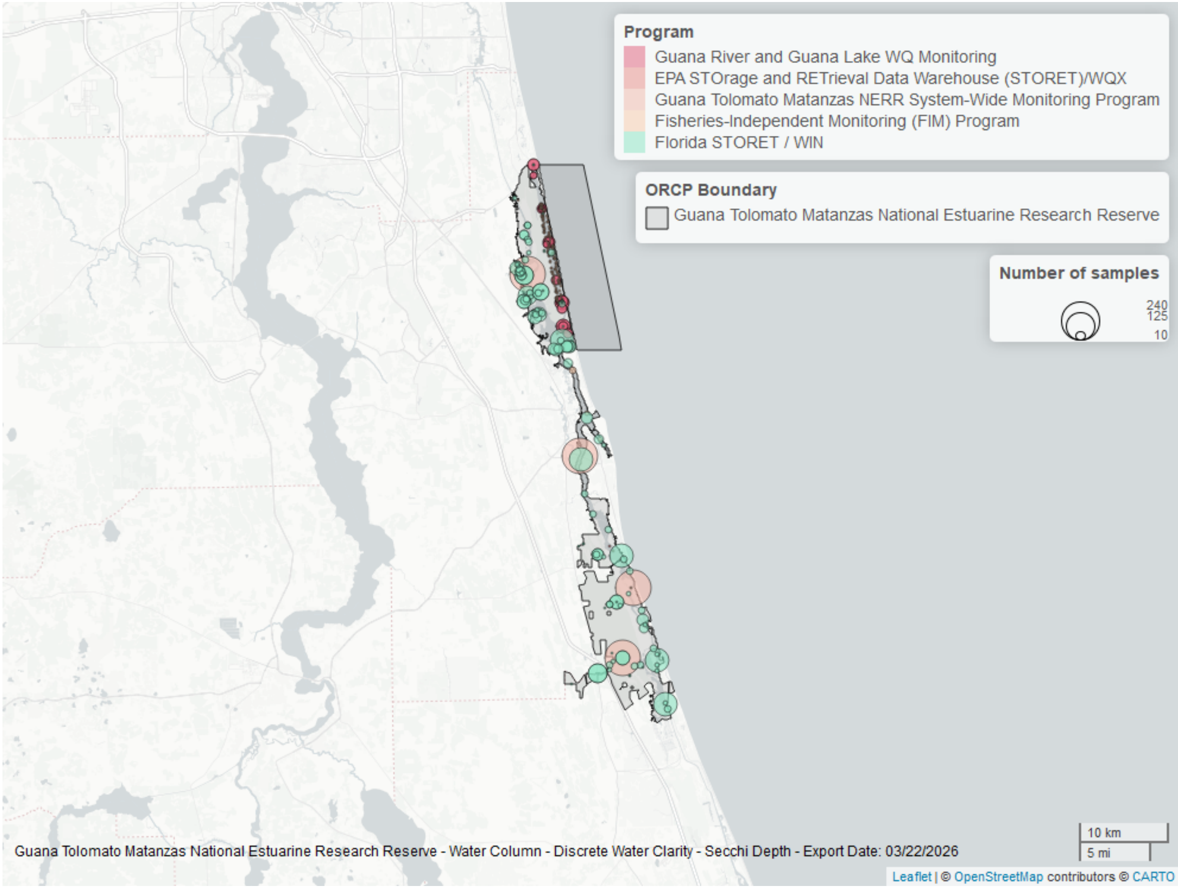

Figure 37: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

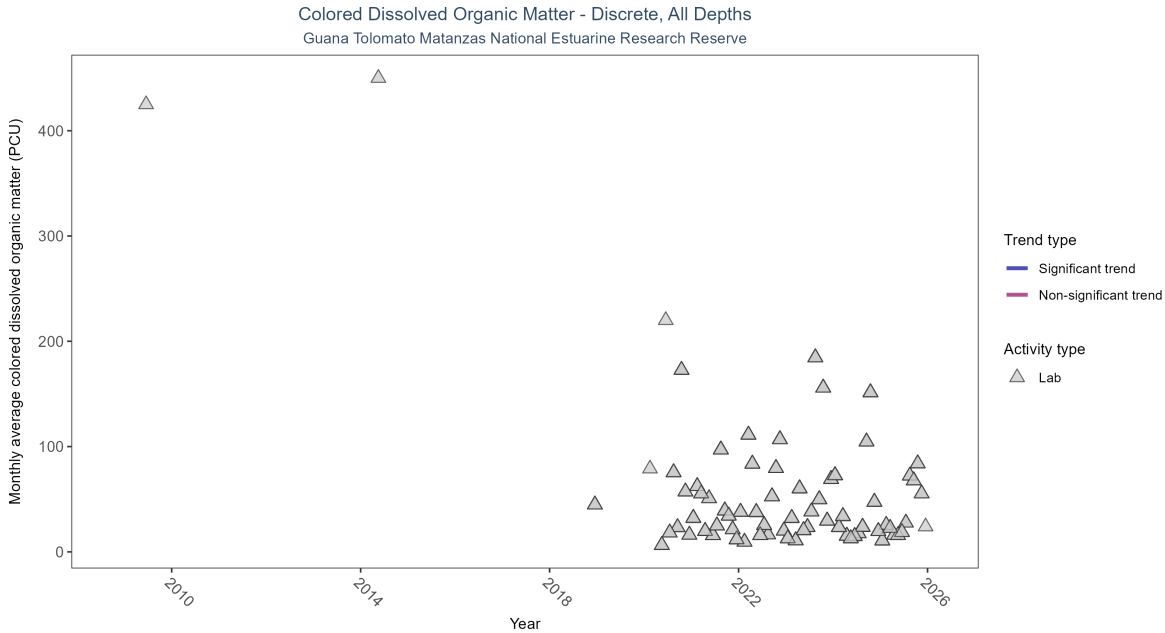

Figure 38: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data to calculate trend	643	9	2009 - 2025	16	-	-	-	-

There was insufficient data to fit a model for colored dissolved organic matter.

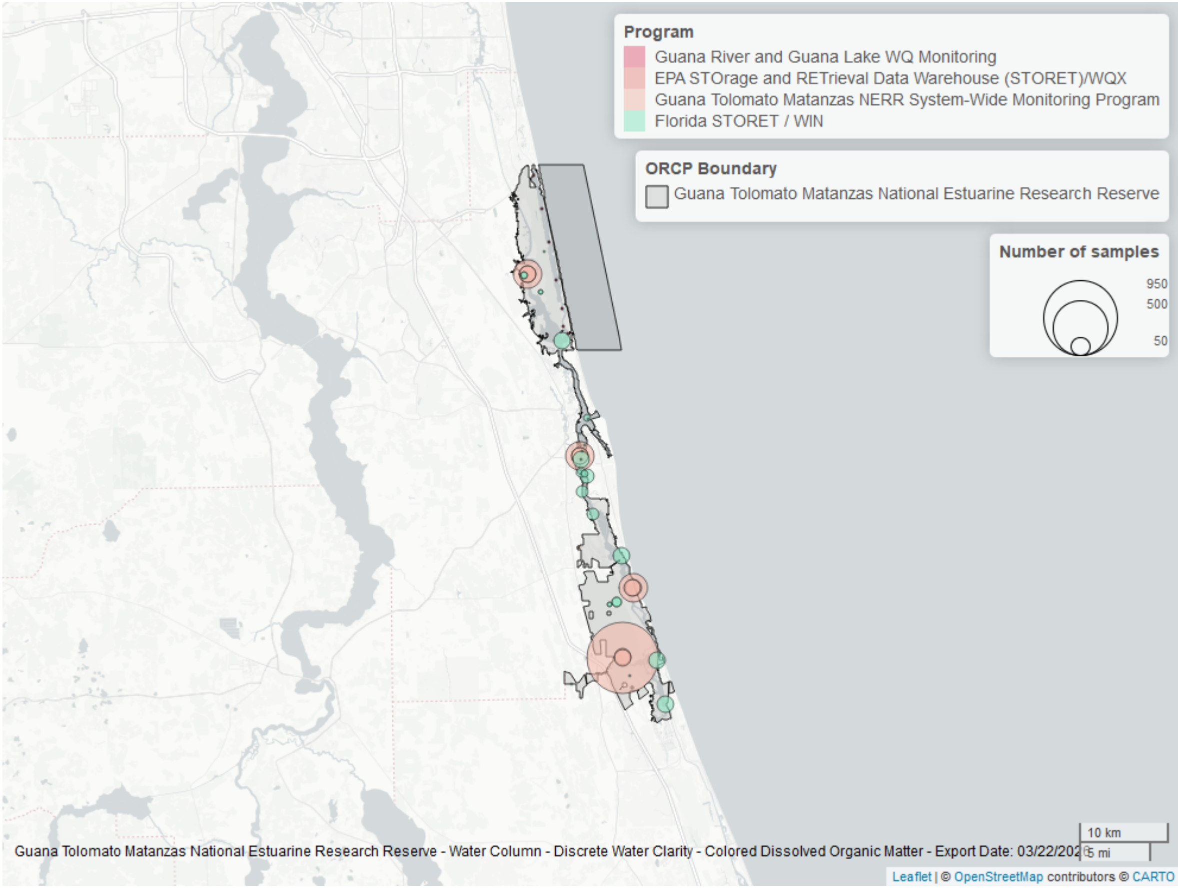

Figure 39: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.